In-Page Navigation Aids for Screen-Reader Users with Automatic Topicalisation and Labelling

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Navigation aids such as headers and internal links provide vital support for screen-reader users on web documents to grasp a document's structure. However, when such navigation aids are unavailable or not appropriately marked up, this situation can cause serious difficulties. This article presents the design and evaluation of a tool for automatically generating navigation aids with headers and internal links for screen readers with topicalisation and labelling algorithms. The proposed tool uses natural language processing techniques to divide a web document into topic segments and label each segment in two cycles based on its content. We conducted an initial user study in the first cycle with eight blind and partially-sighted screen reader users. The evaluation involved tasks with questions answered by participants with information from texts with and without automatically generated headers. The results in the first cycle provided preliminary indicators of performance improvement and cognitive load reduction. The second cycle involved co-designing

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an improved version with two blind experts in web accessibility, resulting in a browser extension which injects automatically generated headers and in-page navigation with internal links, along with improvements in the generation of labels using OpenAI's ChatGPT. The browser extension was evaluated by seven blind participants using the same four texts used to evaluate the preliminary prototype developed in the first cycle. With the two development cycles, the study provided important insights into the design of navigation aids for screen-reader users using natural language processing techniques, including the potential use of generative artificial intelligence for assistive technologies and limitations that need to be explored in future research.

CCS Concepts: • **Human-centered computing → Accessibility technologies**;

Additional Key Words and Phrases: Accessibility, natural language processing, screen readers, topic segmentation and labelling, large language models, assistive technologies

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1 INTRODUCTION

The Web has become a critical platform for accessing information and services. However, the amount of information available on websites can be overwhelming, especially for individuals who use screen readers. Screen readers read the information on the computer screen out loud linearly, making it difficult for users to skip content or read ahead without missing crucial details. This reading method can cause unnecessary cognitive burden, depending on the structure of the content being read [26]. This task can be especially challenging when trying to find specific information, as users may need to remember previously encountered information. If some information needs to be remembered in a lengthy document, users may need to reread entire paragraphs to rediscover what they were looking for [49].

When reading, sighted individuals have various techniques to avoid information overload and save time. They may skim texts, read diagonally, skip sentences, identify crucial elements at a glance, or use visual cues, such as bold font. However, for individuals who are blind or visually impaired, the options are more limited. Users may rely on keyboard shortcuts to skip content, navigate using alternative methods such as headers or link listings, or increase the speed of their screen reader. Nonetheless, people with visual impairments often refrain from skipping content within lengthy texts, as they fear losing important information. Increasing the speed of the screen reader is a possible option, but it comes at the cost of higher cognitive load since a faster information flow demands more attention to comprehend [25].

Experienced users of screen readers often use a strategy of first navigating through all of the headings on a web page to gain a general overview, as noted by Borodin et al. [16]. If they cannot locate the desired information, they may return to the beginning and read through the entire page. With frequently visited websites, experienced users often remember the order of headings and other landmarks, making it easier to navigate quickly to the desired page or section. In categorising different strategies for interacting with screen readers, Power et al. [42] also noted that users often use a "Discovery" strategy to probe headings and other page items to establish the overall structure of a web page.

Research surveys on the habits of screen reader users support the notion that navigating through headings is the preferred method in North America and Europe. A global study involving 1,600 participants [65], mainly from Europe and North America, found that 67.7% of respondents favoured using headers, whereas in Brazil [21], 31.6% of 564 participants opted for this navigation technique.

On the other hand, 35.3% of Brazilian respondents reported using arrow keys to read the content, while 25.2% skipped through the links.

A possible explanation for the regional differences may lie in the prohibitive costs of assistive technologies in Brazil and how alternative, locally-developed technologies were developed. In the case of screen readers, the first system used by many Brazilian blind people was DosVox [15], released in 1993. By 1996, the system already had more than 500 users, which peaked in the following years. Other international screen readers did not have support for Brazilian Portuguese and would cost several months' worth of minimum wage. DosVox had a significant role in widening computer access for blind people in Brazil. However, its first web browser, WebVox, has never made available any feature to provide navigation by headers or other elements. Considering that a vast number of Brazilian blind users learned to use a computer with DosVox, many of them still kept the navigation strategies they learned using such systems, even after the massive move to the free software NVDA, which offers this feature and is now the most widely used screen reader in Brazil. Despite regional differences, both surveys [21, 65] show that different navigation aids are essential for identifying content on lengthy pages.

The significant reliance on page elements for screen reader navigation underscores the criticality of proper HTML markup. Unfortunately, this is only sometimes the case. Incorrectly marked page elements pose a common obstacle for visually impaired individuals when browsing the web [25]. Even following accessibility guidelines, a seamless navigation experience cannot always be guaranteed [39]. Such issues impede using time-saving shortcuts that are vital for blind users.

Several studies have investigated automatic approaches to facilitate text skimming for visually impaired individuals. These studies, including those conducted by Ahmed et al. [2], Gadde and Kankanhalli [25], Giraud and Hautiére [26], and Alves et al. [5], have explored various methods such as automatic summarization [3, 5, 58], filtering redundant content on web pages [25, 26], and simultaneous listening to multiple audio tracks [27]. However, such tools are not widely available for free screen reader technologies used by visually impaired individuals. Furthermore, we have identified a research gap that could potentially improve navigation and information retrieval in lengthy texts by automatically generating headers as landmarks. While organizing information into smaller segments to enhance comprehension is not novel [56], we have yet to find applications of text segmentation and topic labelling for screen readers. Lastly, it is worth noting that these tools are currently not available for the Brazilian Portuguese language.

A previous study [52] by the authors of this aricle explored the automatic generation of headers coupled with screen readers to provide an alternative skimming strategy for visually impaired people when reading long texts in Brazilian Portuguese. That article described an algorithm for segmenting expository text into subtopics and labelling them based on the idea that multiple subtopics may occur during the discussion of one or multiple main topics [28]. Individuals with visual impairments evaluated the proposal, and the results showed that the approach could save time and reduce user fatigue during information-seeking tasks with the aid of automatically generated headers. However, the results also revealed that participants criticized the quality of the labels generated by the previous system version, highlighting the need to improve label generation algorithms and overall accuracy.

Recent advancements in **large language models** (**LLMs**) have expanded the capabilities of **Natural Language Processing** (**NLP**), leading to improved performance and new benchmarks in various NLP tasks. These models can now be applied to tasks such as summarizing text, translating between languages, and generating original content. ChatGPT is an example of LLM, developed in 2022 by OpenAI for open-ended conversations and which is being leveraged for multiple other tasks, in addition to those for which it was designed [1], for example, to explore new opportunities to provide new resources in assistive technologies.

In this article, we present an extension of the previous study [52]. We conducted a second development cycle, which involved eliciting requirements, developing an improved version with two blind experts in web accessibility, building upon the feedback obtained from the user study conducted in the first development cycle, and a second cycle of user evaluations of the implemented browser extension. It resulted in a browser extension that injects automatically generated headers and in-page navigation with internal links, along with improvements in the generation of labels using OpenAI. In this version, the technique focuses on text-based content on web pages. Many web pages contain large amounts of texts, such as in contexts as education, legal information, governmental instructions and others. Evaluations used text-only pages. However, content with other elements could be used, but they would not be considered in the segmentation algorithms. The OpenAI API provides labels of automatically generated textual segments yielded by the segmentation algorithm implemented in the first cycle of this study. Labels generated by this resource are inserted into the text and displayed on the HTML page, along with internal links to the segments, to cater to different user profiles who may use different navigation strategies.

The remainder of this article is organised as follows. Section 2 presents fundamental concepts concerning NLP techniques applied in the study. Section 3 discusses related works covering strategies for providing navigation approaches focused on users with visual impairments. Section 4 presents the implementation and user evaluation of the first development cycle of algorithms for the topicalisation and labelling of text excerpts. Section 5 describes the second development cycle, with the co-design of the browser extension and the improved labelling process. Section 6 presents discussions and limitations. Finally, Section 7 presents conclusions and future work.

2 THEORETICAL BACKGROUND

In this study, we applied NLP techniques to divide lengthy texts into topics and assign them labels based on their content. Word embeddings are frequently employed to comprehend the semantics of the vocabulary. Word embeddings refer to a numeric representation of words that maps each term in a corpus vocabulary to a set of real-valued vectors in a vector space model. These vectors, used to represent each word from a corpus vocabulary, are learned via supervised techniques like sentiment analysis and document classification or unsupervised methods like statistical analysis of documents. These models represent a significant advancement in NLP since they allow for the preservation of a word's semantic, syntactic, and contextual meanings based on its usage in sentences while also being scalable. Many research studies used word embedding for text segmentation [4, 20, 38, 40, 41, 53].

In word embeddings, words with a similar semantic or contextual meaning also have a similar vector representation, while each word in the vocabulary has a unique vector representation. Additionally, the relationship between words, such as verb tenses and gender, is preserved, allowing mathematical operations to be performed on words. For instance, since semantically or contextually similar words share similar representations in a vector space model, in our research, we employed cosine distance between vectors to determine the similarity between words. We employed this method to recognize topic shifts, where the text transitions from one subtopic to another. We assumed that a decrease in similarity indicated a shift in words and, as a result, a change in the subject.

In this study, we employed BERT, a context-based language modelling technique developed by Google AI Language researchers [22]. Its primary advancement involves bidirectional training of language models, allowing for a more comprehensive understanding of language context and flow than previous single-direction language models. Before BERT, language models only examined text sequences from left to right or right to left, resulting in limited context comprehension. For instance, consider these two sentences: (a) I need to go to the *bank* to make a deposit; (b) I went to the river *bank*. A unidirectional encoder would fail in at least one of these examples because the

context for the word *bank* is either on the right (a) or left (b). However, BERT simultaneously uses both left and right context, resolving this ambiguity issue.

Despite the vast amount of available text data, only a few human-labelled training examples are available in task-specific datasets. The lack of human-labelled data represents a problem for deep learning researchers because models require a large amount of data to perform effectively. BERT addresses this data scarcity by training on unannotated web text, generating a general-purpose model known as *pre-training*. By using BERT on a task-specific dataset, it can be *fine-tuned* to create a task-specific model while retaining all of its pre-trained knowledge. This approach simplifies the sharing of BERT models and saves time in research.

Recent progress in Generative Artificial Intelligence includes OpenAI's GPT-3 (Generative Pre-trained Transformer 3) model, which was trained on a much larger dataset, i.e., texts from a vast web corpus, and has demonstrated state-of-the-art performance across various NLP tasks, including translation, question answering, coherent essay writing, and even generating computer programs [31]. ChatGPT, built upon the GPT language model technology, generates human-like text, performs complex tasks, and represents a significant breakthrough in the fields of NLP and AI [50].

One significant role of ChatGPT in the current scenario is enhancing the quality of life for older people and people with disabilities. Kuzdeuov et al. [33] present system can interact with ChatGPT via voice messages and potentially serve as an assistive technology for the visually impaired and blind to access information. As opportunities, Bozkurt et al. [17] suggest that ChatGPT can be used to make education more accessible for students with disabilities, for example, by providing alternative ways to interact with educational content and by automating accessibility features like text-to-speech-and-captioning.

The potential of ChatGPT lies in its ability to increase the accessibility of information. Providing quick responses to inquiries can facilitate access to information and knowledge on diverse topics. Additionally, it can enhance digital inclusion for individuals with disabilities who face difficulties with writing or speaking. With ChatGPT, such individuals can easily communicate and access relevant information.

3 RELATED WORKS

The information available in web applications is structured in HTML [67], according to a tree-based data structure, the DOM, rendered in browsers [66]. Blind users navigate this structure and use different strategies for searching for content, such as reading through the page, navigating through links, using the find feature and navigating through headings or landmarks [24, 65]. Web accessibility guidelines and specifications (e.g., *Web Content Accessibility Guidelines* - WCAG [61] and *Accessible Rich Internet Applications* - ARIA [60]) standardize multiple design strategies for implementing these search strategies in web applications, conveying semantic information in the HTML structure which screen readers can use.

However, the state-of-the-art presents multiple studies reporting that web/mobile developers are not always aware of accessibility specifications [7, 14, 35], leading to a lack of implementation of accessibility requirements in web applications [28, 65], including JavaScript widget libraries [62].

Other studies have also conducted large-scale examinations of web accessibility. The evaluation conducted by WebAim in 2023 of the top 1,000,000 home pages showed an increase in the prevalence of headings on web pages. However, there were fewer than 40% of web pages with headings [64].

These results highlight the importance of implementing automatic adaptation strategies for including accessibility solutions in web applications which do not implement them. In order to adapt the presentation and structure of web applications automatically, different techniques have been used to identify components in web applications so that their markup can be enhanced and made

accessible according to WCAG and ARIA specifications. These studies addressed the use of static JavaScript code analysis for identifying Carousel and Slideshow Widgets [18]; DOM structure changes classification for identifying date picker, drop-down menu, suggestion list, alert window Widets [6, 37] and their sub-components [48]; deep-learning saliency network for selecting regions in images as candidate ARIA landmarks [10]; DOM structure classification of Layout/Relational tables; and DOM structure classification for identifying ARIA landmarks [63]. Unlike state-of-the-art techniques, our research focuses on text analysis technologies to provide heading structures for web applications that do not implement them. Furthermore, navigating through heading elements in a web application is among the most frequently used screen reader search strategies [24, 65].

Some research describes techniques for transcoding for Web accessibility, a category of technologies that aim to transform inaccessible web content into accessible content in real time. The concept involves converting inaccessible web pages on the fly using an intermediary server positioned between the web server and the web browser without requiring content authors to modify their pages. The idea is to convert inaccessible web pages on the fly in an intermediary server between the web server and the web browser without asking the content authors to modify their pages. The papers by Asakawa et al. [9], and Trabzon et al. [57] are discussed in the context of these transcoding technologies, with the former describing them and the latter suggesting modifications for improved user experience. Our study employed NLP techniques to alleviate the difficulties of visually impaired people when browsing poorly structured web pages.

Similarly to our NLP approach, Ahmed et al. [2] reported using automatic text summarization to ease the cognitive load of searching for information in web applications. An efficient way to minimize the cognitive load of screen reader users is to reduce the information to be carried by the software. According to Rayner et al. [43], skimming is a reasonable strategy to cope with the overwhelming amount of text we have to read, given that we are willing to accept a tradeoff between speed and accuracy. Ahmed et al. [2] proposed a method for visually impaired people to skim text. Skimming is a strategy commonly used by sighted people to quickly understand the subject of a given text while searching for specific information. One way of skimming is to skip words in the text, reading only those that catch the reader's attention, reducing the time spent in the text without becoming lost. In the non-visual approach of Ahmed et al. [2], an automatic summarization method was proposed to reduce the content of a text by summarizing every sentence separately. This summary was designed to be independent of a corpus or word frequency to suit better the nature of the internet, which often broadcasts information through short texts.

The algorithm relied on sentences' grammatical and syntactic structure to build a dependence graph. Using this graph and each word's grammatical category, the authors trained a classifier algorithm with human-made summaries to choose which words would have a more significant semantic impact. They noted that most words that humans chose to compose the summary were nouns, verbs, adjectives and adverbs, but in some cases, words could lose meaning without context. Thus, after choosing the most semantically important words, the algorithm uses a tree connection algorithm on the dependence graph to select supplementary words. This approach ensured all selected tree nodes had grammatical relations between them, increasing comprehension of the summarized sentence. The study by Ahmed et al. [2] employed text summarization to enhance the readability of texts within web applications. This research significantly differs from our approach, which segments texts to generate labelled navigation structures in web applications.

Gadde & Bolchini [25] conducted a study aimed at identifying common navigation problems encountered by blind users, identifying which sections are most important at each purchase step on the Amazon website, and developed a tool that can suggest user sections based on an importance ranking. Gadde & Bolchini [25] used a crowd-sourcing approach to rank page sections according to their importance. This study analysed the Amazon online store, and the importance of each

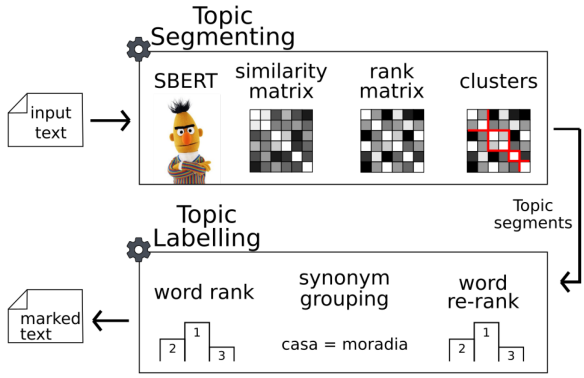


Fig. 1. Algorithm framework.

section was based on its weight towards the buying decision. Based on this ranking system, the user could use the proposed tool to jump directly to specific sections according to their importance, making navigation more straightforward and practical.

In order to aid the navigation of screen reader users, Giraud et al. [26] proposed a web content filtering approach. A tool was developed with features to filter redundant and irrelevant information for the user's tasks, such as ads. Redundant information is presented more than once; for instance, a website logo or a menu repeated on different pages, and irrelevant information adds nothing to the user's tasks. In that approach, users hear all the relevant information when they enter a site. However, redundant information is ignored in the following actions during the same session (such as when the user goes to another site page).

Other studies have also investigated ways of improving the navigation for people with visual impairments in different contexts, such as navigating in content with Chemistry notation [32]. Barbosa et al. [11] proposed a browser extension to enable command-based interaction to standard website features, such as login, signing up and others. The evaluation showed how the commands may help navigate otherwise difficult websites. However, they also showed that users would need human support in some situations, even with automated support.

In this study, we investigated the use of transformer-based models for segmenting unstructured text content and generating heading structures for web applications. The generated heading structures can be applied to any web application and are not domain-specific, unlike the crowdsourcing strategy proposed by Gadde & Bolchini [25]. Our strategy focuses on generating and adding navigation mechanisms to the web application rather than removing or simplifying existing mechanisms and content. This work extended our previously reported prototype by implementing the heading generation strategy in a browser extension using ChatGPT.

4 FIRST CYCLE—AUTOMATIC HEADER GENERATION

The main goal of this study was to automatically create headings within lengthy texts to assist visually impaired individuals in locating information more easily. By inserting headers inside long texts, users could infer the content of a section by reading the header or use it to map where each information is located in the text mentally.

Figure 1 illustrates the framework of our algorithm. The algorithm employs SBERT [44], a sentence processing framework that utilizes BERT, to compare each sentence of the input document and generate a similarity matrix. A filter is then applied to the similarity matrix to create a ranking matrix. The algorithm uses this ranking matrix to identify topic shifts in the document, dividing

the text into contiguous sentence clusters called topic segments. The topic labelling algorithm processes each topic segment separately, ranking words and grouping synonyms to generate headers while avoiding redundant words automatically. This process results in a segmented text labelled with headers that can be easily adapted for other purposes, such as HTML injection or screen reader tags. Further details on this process are available in the following sections.

The current version of the algorithm only analyzes textual information on web pages. Information in images, tables and other interactive elements such as forms and buttons are not currently analyzed.

4.1 Automatic Topic Segmentation Algorithm

The segmentation algorithm developed was based on ToSe-Word2vec [40], which combines word embeddings with C99 [19]. We used BERT, a state-of-the-art model, to calculate similarities in this study. BERT was chosen over other options to observe how this new model would perform in this classic algorithm. The algorithm receives a list of sentences parsed with NLTK[1]—a NLP toolkit - and produces a marked file with sentences divided by topic. It is important to note that any indication of paragraph boundaries is lost during the parsing process. Consequently, paragraphs may be split if the algorithm identifies a topic boundary between two of their sentences.

For a text of n sentences, a similarity matrix $n \times n$ is created using SBERT,[2] a python framework that creates sentence embeddings using a BERT model. We used a multilingual BERT model[3] [45] to enable comparing the segmentation results with previous research, which was readily available for usage through the SBERT library. The similarity is calculated using cosine distance over the embedding of two sentences being compared. Lastly, topic boundaries are detected by clustering sentences using Reynar's maximisation algorithm [47]. It divides the matrix into clusters when it detects a high similarity difference between two sentences.

Figure 2 illustrates this process. Initially, the entire rank matrix is considered a coherent text segment. Lighter colours represent high similarity. The algorithm identifies a boundary between the second and third sentences, indicated by a darker colour, signifying a low similarity between these sentences. Since only this boundary was detected in Step 1, the third sentence onwards is considered a coherent segment. The same happens in Step 2, forming another cluster with sentences 3 and 4. Step 3 finishes the process by detecting the last boundary, resulting in four text segments detected for this example.

4.2 Automatic Topic Labelling Algorithm with BERT

In our labelling process, we assign keywords to each topic segment separately and rank the words based on frequency. The idea behind this approach is that if a word appears repeatedly in a text fragment, it is likely related to the topic being discussed. We first remove stopwords[4] to ensure the label consists of only meaningful words. We started the stopword list with the default from NLTK and added other words that added little value for labels. Additionally, we leverage the insights from Ahmed et al. [2] that nouns, verbs, and adjectives are the most semantically significant words in a sentence. To filter out other types of words, we use part-of-speech tagging. Since NLTK lacks a part-of-speech tagger for Portuguese, we used UDPipe [54], a trainable pipeline for tokenization, tagging, lemmatization and dependency parsing of CoNLL-U files.

[1] Available at https://www.nltk.org/
[2] Available at https://www.sbert.net/
[3] Available at https://huggingface.co/sentence-transformers/distiluse-base-multilingual-cased-v1
[4] Stopwords are the most common words in any language and do not add much information to the text. There is no universal stopword list.

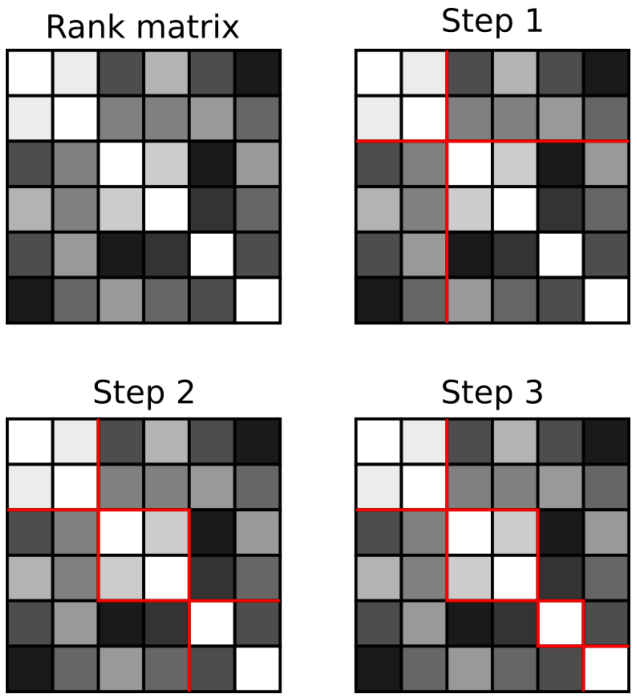


Fig. 2. Example of the clustering algorithm.

After ranking words based on frequency, each word is grouped with its synonym sets (synsets) to prevent the selection of semantically similar words as labels, which could lead to redundant information. Each word in the text is lemmatized using UDPipe. The lemmas are then used to create a synset by combining all of its synonyms from wordNet, available in both NLTK and TEP.2.0[5] [36]. We used two sources for this step to complement the other when it lacks a rare word or includes only a few synonyms.

Our labelling algorithm generates labels with varying keywords due to the automatic segmentation step, which creates topics of different sizes. When a topic is too small, the number of keywords used to label it may be too high, resulting in irrelevant words being included. Conversely, a large topic may not have enough keywords to encompass all of its subjects. To address this issue, we decided to use various keywords in our labels that would suit both scenarios. We tested the label size ad-hoc on the CSTNews corpus[6] and determined that topic labels should vary from 3 to 5 keywords based on the word count of each text segment. This building ensures that larger segments have enough words to describe them, while smaller segments do not include irrelevant words due to their low word count.

In Figure 3, we present an example of a text snippet about Queen Elizabeth II from Wikipedia, which has been segmented into topics and labelled using our algorithm. This example is composed of 14 sentences and three paragraphs. The first topic label (part a in the text): *elizabeth, independent, state, queen, kingdom*) It has more words because it is long. We believe this helps the user have a better understanding of its content. The second topic (part b) is too short to rank five keywords.

[5]TEP is a database that stores sets of synonyms and antonyms to each other.
[6]A corpus of news texts in Brazilian Portuguese available at: https://sites.icmc.usp.br/taspardo/sucinto/cstnews.html

Text in Portuguese	Text in English
(a) *ISABEL INDEPENDENTE ESTADO RAINHA REINO*	(a) *ELIZABETH INDEPENDENT STATE QUEEN KINGDOM*
Isabel II ou **Elizabeth** II (Londres, 21 de abril de 1926) é a atual **Rainha** do **Reino** Unido e de quinze outros **estados independentes** conhecidos como **Reinos** da Comunidade de Nações, além de chefe da Commonwealth formada por 53 **estados**. Ela é a primeira monarca feminina **soberana** da Casa de Windsor e a Governadora Suprema da Igreja da Inglaterra. Em algum de seus outros **Estados soberanos**, ela possui o título de Defensora da Fé. **Isabel** se tornou a Chefe da Comunidade Britânica quando assumiu o trono em 6 de fevereiro de 1952, após a morte de seu pai, o rei Jorge VI. **Isabel** é a **rainha** de quatro países **independentes**: **Reino** Unido, Canadá, Austrália e Nova Zelândia. Entre 1956 e 1992 o número de países do **reino** britânico variou já que certos territórios ganharam sua **independência** e outros tornaram-se repúblicas. Atualmente, além dos quatro primeiros **estados** mencionados, **Isabel** é **rainha** da Jamaica, Barbados, Bahamas, Granada, Papua-Nova Guiné, Ilhas Salomão, Tuvalu, Santa Lúcia, São Vicente e Granadinas, Belize, Antígua e Barbuda e São Cristóvão e Nevis. **Isabel** nasceu em Londres e foi educada particularmente em casa. Após a Crise da abdicação de Eduardo VIII, tio de **Elizabeth**, houve a ascensão de George VI ao trono, o que fez de sua primogênita a herdeira presuntiva da coroa. **Isabel** passou a assumir deveres públicos durante a Segunda Guerra Mundial, em que ela serviu no Serviço Territorial Auxiliar.	**Isabel** II or **Elizabeth** II (London, April 21, 1926) is the current **Queen** of the United **Kingdom** and fifteen other **independent states** known as **Kingdoms** of the Commonwealth of Nations, as well as the head of the Commonwealth consisting of 53 **states**. She is the first female **sovereign** monarch of the House of Windsor and the Supreme Governor of the Church of England. In some of her other **sovereign states**, she holds the title of Defender of the Faith. **Isabel** became the Head of the Commonwealth when she ascended to the throne on February 6, 1952, following the death of her father, King George VI. **Isabel** is the **queen** of four **independent** countries: the United Kingdom, Canada, Australia, and New Zealand. Between 1956 and 1992, the number of countries in the British realm varied as certain territories gained their **independence**, and others became republics. Currently, in addition to the first four **states** mentioned, **Isabel** is the **queen** of Jamaica, Barbados, Bahamas, Grenada, Papua New Guinea, Solomon Islands, Tuvalu, Saint Lucia, Saint Vincent and the Grenadines, Belize, Antigua and Barbuda, and Saint Kitts and Nevis. **Isabel** was born in London and received private education at home. After the abdication crisis of Edward VIII, **Elizabeth**'s uncle, there was the accession of George VI to the throne, making his eldest daughter the presumptive heir to the crown. **Isabel** began to assume public duties during World War II, in which she served in the Auxiliary Territorial Service.
(b) *ANO CASAR PRÍNCIPE*	(b) *YEAR MARRY PRINCE*
Ela se **casou** com o **príncipe** Filipe da Grécia e Dinamarca em 1947, com quem teve quatro filhos: Carlos, Ana, André e Eduardo. Seu pai morreu em fevereiro de 1952 e Isabel ascendeu ao trono aos 25 **anos**. Sua coroação ocorreu no **ano** seguinte e foi a primeira a ser televisionada.	She **married Prince** Philip of Greece and Denmark in 1947, with whom she had four children: Charles, Anne, Andrew, and Edward. Her father died in February 1952, and Isabel ascended to the throne when she was 25 **years** old. Her coronation took place the following **year** and was the first to be televised.

Fig. 3. Examples of labels generated by the tool.

Table 1. Time Taken to Segment Each Text

Text	Time(s)	Sentences	Words
Text 1	1.5	26	720
Text 2	1.26	28	748
Text 3	4.4	49	955
Text 4	10.47	82	1131

Table 2. Time Taken to Label Each Text

Text	Time(s)	Topics	Words
Text 1	1.16	6	720
Text 2	1.13	6	748
Text 3	2.5	5	955
Text 4	3.21	7	1131

Thus, only three are used (*year, marry, prince*). After the third word in the heading, the remaining words follow the order in which they appeared in the topic, as there were no other repetitions.

The initial segment consists of 10 sentences and almost three paragraphs from the original. The last words of each paragraph from the original text are underlined; it is possible to notice that the two first and half of the third were included in the first segment. It describes who Isabel II is, which countries she is the queen of and a bit about her early life before the coronation.

The second paragraph, on the other hand, is brief and consists of only three sentences. Remarkably, the second segment concludes with the same sentence that concludes the original paragraph. It discusses Isabel II's marriage to Prince Philip, their children, her father's death and her coronation.

The header words are highlighted in bold in the given paragraph. For the first label, the word "Isabel" was counted with eight repetitions, which includes the variations of "Elizabeth". The words "Independent", "independence", and "sovereign" were counted for "Independent" with five repetitions. Similarly, the word "States' was counted for "state" with four repetitions. The word "queen" appeared three times, and the word "Kingdoms" was counted towards "Kingdom", repeated twice.

For the second topic, only the word "year" was repeated twice, considering both "year" and "years". The words "married" and "prince" occurred only once, counted towards the labels "marry" and "prince". It should be noted that the verb "marry" appears in its infinitive form due to lemmatization, whereas in the text segment, it appears in the past tense as "married".

4.3 Algorithm Results

The processing time in seconds for segmenting each text in the user study is shown in Table 1. A clear correlation between the number of sentences and processing time can be observed. Texts 1 and 2, with similar numbers of sentences, have almost identical processing times, while Text 3, with nearly double the number of sentences, takes three times as long. Similarly, comparing Text 3 to Text 4, which has nearly twice as many sentences, yields the same result with both processing time and sentences doubled. This outcome was expected since the most time-consuming step in the algorithm is detecting topic boundaries between sentences in the rank matrix, which increases in size with the number of sentences.

In the case of the labelling algorithm (Table 2), the number of words increases the processing time. It is possible to observe that, even though Text 3 has fewer topics than Texts 1 and 2, the time increases according to the number of words, and the same happens with Text 4. This result was

Table 3. Error Rates—Performance Comparison with Previous Algorithms

Algorithms/Topic length (in number of sentences)	3–5	6–8	9–11	3–11
Our Algorithm	27%	12%	8%	15%
C99	18%	10%	10%	13%
Reynar	21%	18%	16%	22%
Misra	16%	11%	11%	12%
TextTiling	44%	43%	48%	46%

also expected since the algorithm has to rank every word for each topic and create synonym sets to group words.

Considering the aim of this tool is to be used in texts even longer than the ones in the user study, much improvement is still needed to decrease processing time. For a text with just 1131 words, waiting 14 seconds to have automatically generated headers could discourage the user from using this tool. Conducting a user test with the algorithm implemented as a plugin would be necessary to confirm the impact of waiting time on usability.

Using a multilingual BERT model allowed us to compare our results with previous research in English. We tested our algorithm with Choi's dataset[7] [19] and measured error rate (P_k) [12]. According to the P_k metric, an algorithm that assigns all boundaries correctly receives a score of 0. At each location, the algorithm determines whether the two ends of the probe are in the same or different segments in the reference segmentation and increases a counter if the algorithm's segmentation disagrees.

Table 3 compares the results we achieved with our algorithm and previous research. The numbers in each column's header indicate the size of topics; 3–5 means topics were 3 to 5 sentences long. The table shows that C99 and Misra [38] still outperform our algorithm in most scenarios, except for the scenario with the longest documents (9–11). Still, we detected boundaries with fewer errors than Reynar's [46] and TextTiling [28]. Our solution is more accurate in identifying boundaries in longer texts. This result could be because the vocabulary variation in shorter texts is not as significant, resulting in fewer subtopic variations and, consequently, fewer topical boundaries. Although we based our algorithm on Naili's adaptation of C99 to use Word2Vec, it was impossible to compare results since the corpus they used was different and not made public.

The output of the automatic header generation algorithm is a text file with generated headers followed by their respective text segment. Each word from the header is separated by a space and is located right above its topic segment. An empty line separates topics.

4.4 First Cycle—User Study Design

The study was conducted remotely due to the COVID-19 pandemic. It involved evaluations of different web pages with four texts in two scenarios in a mixed design:

(i) Without headers/ With headers: Texts 1 and 2 are presented in their original form, without headers, while texts 3 and 4 are presented with automatically generated headers by our algorithm;
(ii) With headers/ Without headers: Texts 1 and 2 are presented with automatically generated headers by our algorithm, while texts 3 and 4 are in their original form, without headers.

[7] Available in http://web.archive.org/web/20010422042459 http://www.cs.man.ac.uk/Ûchoif/software/C99-1.2-release.tgz

Table 4. Participants' Demographic Data

Code	Age	Gender	Sight condition	Years using screen readers
P1-C1	26	F	Total blindness	15
P2-C1	33	M	Total blindness	10
P3-C1	51	M	Residual light vision	20
P4-C1	43	M	Movement perception.	13
P5-C1	28	M	Total blindness	12
P6-C1	22	M	Residual light vision	16
P7-C1	68	M	Total blindness	18
P8-C1	28	F	Total blindness	20

Participants were allocated for each group in turns when they volunteered, i.e., if a person were assigned to scenario (i), the following volunteer would be assigned to scenario (ii). Participants answered three questions for each text before advancing to the subsequent text. If participants could not find an answer, they could skip it. After each text, participants rated the cognitive load used to complete the task, based on Overall Workload [29]. The tests, including consent terms and demographic questions, were available via Google Forms. Each session was recorded via Google Meet. The recordings were used to analyze users' interaction with long texts in both scenarios and identify possible problems.

The study was approved by the Federal University of Lavras Research Ethics Committee, with CAAE 45372321.3.0000.5148. Individuals had to be at least 18 years old and have completed high school to participate. As a result of the Covid-19 pandemic, all assessments were conducted remotely. Since the researchers could not provide in-person technical assistance, all participants needed to have basic experience with screen readers and computers to complete the tasks. Participants were recruited by contacting people who participated in previous studies in the ALCANCE laboratory (Research Lab on Accessibility, Usability and Computational Linguistics) and posting an invitation for the study in Facebook groups aimed at visually impaired users. Table 4 displays the participants' demographic information.

4.5 Test Procedure

The tasks comprised three questions and a text that could either have no headers or the automatically generated headers according to the scenario the participant was assigned. They were allowed to interact with the text how they preferred, with whichever voice speed, shortcuts or other techniques they were accustomed. Participants were free to skip questions if they were unable to answer them, but everyone answered them.

After each task, participants evaluated the level of cognitive load they felt was necessary to complete the task. We used a scale from 1 (very low cognitive load) to 10 (very high cognitive load), based on Overall Workload [29]. We favoured this scale because we wanted a simple way to measure mental effort for each task. We knew other alternatives to test cognitive load, such as the NASA-TLX [55] which separates the total cognitive load into six subscales. However, we chose a more straightforward option to avoid tiredness that would impact the results of the following tasks. Hill et al. [29] shows Overall Workload is easier and faster to complete than the NASA-TLX. Van Gog et al. [59] also showed evidence that repeatedly measuring mental effort after each task in the series is preferable to rating only once or retrospectively.

At the end of the test, questions were posed in a guided conversation to acquire feedback about the proposed algorithm. We preferred the conversation after the test because a method like

think-aloud, where the user gives feedback while testing, would impact both the time and cognitive load measurement.

4.6 Prototype Evaluated

The user study prototype consisted of web pages created using the outcome of two algorithms. The first algorithm detected topic boundaries and divided the text accordingly, while the second algorithm automatically labeled these topic segments by generating keywords from each text segment.

We chose four informative texts to evaluate the proposed algorithm and used the automated division and labeling of topics. These topics were then automatically saved to a text file. Since the test was conducted remotely, we needed to find a way to make the text available online while maintaining its readability, including separating paragraphs with headers. To accomplish this, we used Google Forms that allowed us to save and share the results, and its typical screen reader shortcuts worked well.

We simulated the proposed text result of our proposed algorithm inside Google Forms. Each topic was inserted in a separate text box to allow the use of both header and paragraph navigation since Google Forms does not allow for HTML editing. Each header is a *title* for a text box, and the segment content is its *description*. We tested this setup using NVDA to ensure that common shortcuts would work as expected.

4.7 Data Analysis

We conducted a descriptive statistical analysis on cognitive load and time taken for task completion. Time was not considered when the study was disturbed by an external problem, such as the screen reader losing focus on the web page. We applied a within-subject non-parametric Signed-Rank Wilcoxon test to assess whether there was a significant difference between texts with automatically generated headers and without headers for each participant.

We also conducted a qualitative analysis on the feedback collected during post-test guided interview, identifying positive and negative aspects of the proposed tool and situations that caused positive or negative reactions.

4.8 User Study Results

This section presents the results obtained in the user study. Descriptive data for the time taken and cognitive load was measured to compare scenarios with and without automatically generated headers. Qualitative data was also obtained via a guided conversation with participants after finishing all tasks. We present their statements about the proposed tool, separated by topics.

4.8.1 Cognitive Load. We measured cognitive load to investigate if separating a long text into smaller topics labelled with automatically generated headers would reduce the mental effort for users. Based on the Overall Workload [29] measurement, a 10-point scale was used to measure cognitive load from 1–very low cognitive load to 10–very high cognitive load. This scale enabled measuring cognitive load with lower intervals right after each task without excessively increasing the total testing time.

Table 5 shows the cognitive load scores for each participant to complete tasks on each text. Texts with automatically generated headers are indicated with (h).

Since each scenario (with or without headers) had two texts, the average between each text for each scenario was calculated to verify whether automatically generated headers helped users navigate and find information inside texts.

Table 5. Cognitive Load Scores

Participant	Text 1	Text 2	Text 3	Text 4	w/h	n/h
P1-C1	4 (h)	3 (h)	6	8	3.5	7
P3-C1	2 (h)	2 (h)	3	3	4.5	5.5
P5-C1	7 (h)	7 (h)	9	5	2	3
P7-C1	1 (h)	5 (h)	8	6	4	3.5
P2-C1	5	6	4 (h)	5 (h)	7	7
P4-C1	3	4	5 (h)	3 (h)	4.5	6
P6-C1	8	4	6 (h)	3 (h)	3	7
P8-C1	8	9	6 (h)	5 (h)	5.5	8.5

w/h = with headers; n/h = no headers.

Table 6. Time Taken on Each Task

Participant	Text 1	Text 2	Text 3	Text 4	w/h	n/h
P1-C1	539 (h)	402 (h)	494	632	471	563
P3-C1	531 (h)	451 (h)	628	507	473	568
P5-C1	910 (h)	572 (h)	959	424	741	692
P7-C1	681 (h)	878 (h)	1418	872	780	1145
P2-C1	437	454	640 (h)	708 (h)	674	446
P4-C1	851	473	432 (h)	504 (h)	468	662
P6-C1	744	574	643 (h)	640 (h)	642	659
P8-C1	1570	1550	1069 (h)	1166 (h)	1118	1560

w/h = with headers; n/h = no headers.

4.8.2 Time on Task. The time for each task was measured from the moment the screen reader started reading the title until the participant clicked to enter the next page. Time taken with problems was deducted, i.e., screen reader losing focus from the web-page or the participant having to look for a battery charger. Time taken in each task was expected to vary between users since they were allowed to use their preferred speed. Table 6 shows the time taken in seconds for each participant to complete tasks on each text. Texts with automatically generated headers are indicated with (h).

4.9 Participants' Perception on the Use of Automatically-Generated Headers for Navigation

4.9.1 Usage of the Headers. Four out of eight participants reported that the headers facilitated the completion of tasks. This fact can be attributed to the segmentation of the text by topics and how the labels helped to know where participants were or infer the text's subject. The desired experience was achieved for some users, as Participant 2 reported "It was fantastic when I got used to it because I felt like running my eyes over the text.", almost like skimming it.

For other users, the headers were helpful during a second read, helping locate the desired text segment. Participant P5-C1 said *"It helps more for a second read when I've read it already, rather than in the first read"*. Participant P7-C1 agreed the headers were helpful and said this tool would be even more useful for longer texts:

> *I certainly preferred [the texts] with markers [headers]. And these texts weren't even that long. If they were longer it would be harder to remember where information was without the headers.*

In most cases, participants who did not use the generated headers explained that they are used with line by line or paragraph reading to read the texts thoroughly and then answer the questions. As Participant P8-C1 reported: *"In my case, it wasn't useful because I read it line by line."* However, even in this group, some participants considered the tool useful. As participant P4-C1 said:

> *"I didn't use the headers. I read paragraph by paragraph but if needed I could go back in the text at a certain point, headers would help."*

Participant P6-C1 also said they did not use headers, but argued:

> *"It facilitated locating myself with the screen reader. In the texts with no headers I had to read line by line. In the ones with headers I just continued reading the text and could go back to read afterwards."*

4.9.2 Infer Topic Content from Header. Six out of eight participants reported being able to infer topic content based on the header. Participants P5-C1 and P6-C1 said they could not know the subject of a topic when reading the automatically generated header. Participant P5-C1 reported "*the headers are a grouping of words that don't hold meaning.*" and that *"the words seemed disconnected"*. Participant P6-C1 said the headers hinted at what the segment was about but was not enough to know for sure. He also added that the text became easier to read:

> *"It is noticeable that it helps finding information but I didn't use much. The words used in the headers can give us a hint, still, there is some doubt left. In the text without headers the impression is that it will never end. [...] Having the headers, with these highlighted words makes it less exhausting."*

Participant P6-C1 also added headers could still improve in the choice of words and their presentation:

> *"I think the algorithm can still improve on the choice of words. The words in the headers were from the text segment but the way they were placed, thrown in, could improve."*.

The participants with positive responses said the labels were enough to guess somewhat the subject of its segment. Participant P2-C1 said *"it helped to find what I was searching."* Participant P8-C1 stated it was similar to searching for keywords: *"I think the headers were good according to what the text had to offer. Because many searches we do, we use keywords."*

4.9.3 Navigation. All participants said the topic separation helped them navigate the text. Participant 1 answered positively, saying *"It was like navigating section by section"*, reminding the topic separations seen in Frequently Asked Questions of websites.

Participants P4-C1, P5-C1, and P7-C1 also reported that the headers helped them remember where something is and find it again. Participant P5-C1 said the tool was a positive surprise because, at first, they were not convinced about it, but after using it, helped to locate themselves when navigating the text for question answering:

> *"I started thinking it [the headers] was unnecessary, [...] but by simply having them it created a checkpoint for me. [...] It helps me to fragment the text in my mental reading. I can get used to memorizing where something is by its header."*

Participant P6-C1 said the headers helped navigating for another reason, enabling him to use the header listing shortcut to navigate:

> *"It helped a lot because I could use shift+H to navigate through headers, it was much faster. It gives me agility."*

4.9.4 Desire to use the Tool. All participants said they would use a tool like this if available. Participant P6-C1 stated it would be a useful tool since *"There are many sites that don't even have headers."*. Participant P5-C1 was eager to use an algorithm like this for his work: *"Yes, mainly for my reading activity, proofreading."*. Participant P2-C1 also commented on a problem they often face when reading long texts with no headers:

> *"When there is no marking, navigating takes more time because we have no anchor or reference to use with the screen reader."*

Two participants reported that the machine did not seem to have done the segmentation and that they felt like topics were breaking where they should, i.e., at the end of paragraphs. However, it is worth noting that when the segmenting algorithm processes the input text, it has no paragraph information; it simply processes a list of sentences. Participant P2-C1 said:

> *"In most cases I didn't notice it was made by machine. In some points I thought 'this is a little weird' because the subject continued in the next paragraph."*

Participant P5-C1 also reported something similar:

> *"In some cases I felt like the topics were too much divided by paragraphs, and not by subject. The algorithm would do 'this paragraph is this block, this other paragraph is that block'. [...] I felt like the topic breaks were actually the paragraphs, just like someone had intentionally ended one and started the next. If this was the algorithm, it's very good."*

Because of these statements, we checked the number of paragraphs from the original text that matched the topic boundaries detected by our algorithm. Text 1 had five boundaries; four were at the end of a paragraph. In text 2, all boundaries matched the ends of paragraphs. For text 3, three boundaries out of four matched paragraph breaks. In text 4, just two boundaries out of 6 matched the end of a paragraph. This fact might indicate that, even in cases where the boundaries were not in the same place as the paragraph ending, it still made enough sense for the participant to think it was a natural break.

5 SECOND CYCLE—BROWSER EXTENSION WITH OPENAI

The first development cycle and user evaluation provided valuable insights into how to develop a browser extension to provide navigation aids to people using screen readers. One of the main aspects raised in the evaluations was that there were limitations to the quality of the algorithm-generated labels. Half of the participants reported not using the headers when completing the tasks. This result indicated that although users saw the benefit of using the proposed tool, the header labels still needed improvement. The evaluations also showed that not all users were proficient enough to use navigation with headers efficiently. We identified that the proposed approach for automated text segmentation and topicalisation was promising from the results obtained in the first cycle. However, better language models were needed to improve the generation of headings. The study also showed the need to explore in more detail how users with visual impairment would use such tools in texts in their web browsers.

Following the input from the first cycle, the second cycle involved the requirements elicitation process with two blind experts, using the OpenAI API to create improved labels for the segments generated by the topicalisation algorithm and a second cycle of user evaluations.

With the recent rise of advanced language models like ChatGPT, powered by OpenAI, there has been a heightened interest in leveraging these models for tasks such as text classification, question answering and machine translation [51]. A language model is an artificial intelligence system that learns to generate or process text based on training examples. They are trained on large amounts of text to learn to predict the next word or phrase based on the previous context. They can also be used to find topics in a text by analyzing the words and phrases used in the text and identifying patterns and clusters of related words. The works have used language models to help automatically identify topics in texts, such as in social media on electric vehicles [30] and in Spanish plenary sessions [23]. However, language models typically do not provide explicit labels or tags for these topics.

ChatGPT leverages deep learning to generate text (including essays, stories, poems and code), enabling performing diverse tasks without specific training [51]. As an AI language model, ChatGPT can generate topics in a text by analysing the words, phrases, and concepts contained within it. From this, we believe that ChatGPT can determine which words are most relevant to each topic segment in a text. The idea is to compare the new labels with those generated by our first algorithm that considered the word ranking for each topic segment.

5.1 Requirements Elicitation Process with Blind Experts

The development of the browser extension in the second cycle was based on the feedback from the user evaluation of the prototype in the first cycle and the involvement of two blind experts in a co-creation process.

The two blind experts (co-authors of this paper) were (1) an MSc student of Informatics in Education with a degree in Information Systems, with several research papers published in topics related to web and mobile accessibility; (2) a researcher with an MSc and BSc degree in Communication studies, who has worked in IT support in an Assistive Technology Centre, and has also published several studies related to digital accessibility.

The browser extension development encompassed several sessions involving blind experts and other researchers to discuss the results from the user evaluation in the first cycle and possible solutions. The two blind experts provided critical insights to define the interaction strategies used and to define how the ideas brought by other researchers from NLP and Web technologies could be applied in practice.

Important design implications were derived in the cocreation process, which were reflected in the browser extension:

(1) **Different navigation strategies:** The blind experts gave practical examples of how people with different levels of expertise used different strategies to navigate lengthy texts. Despite the initial perception of the importance of providing only headers, they added that other strategies should be provided to cater for people with different levels of expertise with screen readers, especially in the case of Brazil, as a report from a Brazilian survey [21] shows different behaviours in comparison to people in North America and Europe [65]. They recommended that the labels generated for headers also be made available as internal links at the beginning of the document.
(2) **Keyboard actions:** The experts provided insights into keyboard shortcuts to be used in the extension. They recommended the keys to activate the extension be easy to reach for users.
(3) **Improving the quality of labels:** When analyzing the results from the first cycle of development and user evaluation, the experts agreed that slicing the text with some keywords indicating the topic could help to some extent in performing their tasks. However, improved labels were essential to decide whether the content they sought was at that

point. The two blind experts agreed with the suggestions given by other researchers to explore the use of the Open AI API, as suggested in other studies, as a potential tool to implement features that could act along assistive technologies [17, 33].

(4) **Navigation with internal links:** The experts indicated that internal links to the beginning of the identified text segments should be placed at the beginning of the document. Users using this feature could be more used to having keyboard shortcuts and links such as "Skip to content," usually placed at the beginning of web pages.

5.2 Browser Extension Development

Following the requirements elicitation process, the development involved several iterations by implementing working versions and obtaining feedback from the blind experts.

The browser extension was developed using JavaScript for Google Chrome. The implementation utilizes a server-based approach for the text topic segmentation using BERT, as was done in the first cycle.

The step-by-step process of the browser extension works as follows:

(1) Users activate the browser plugin using a keyboard shortcut;
(2) The browser extension reads through the page to obtain all textual content and loads a cleaned up version of the text in the client computer's memory, which is sent to the server;
(3) The server runs the topic segmenter algorithm using BERT to analyze what texts segments are present in the text, as described in Section 4;
(4) The client receives the segments identified by the topic segmenter. For each segment, the browser extention makes a request using the OpenAI API to ChatGPT version 3 with the command "Create a header in pure text for the following text …".
(5) The browser extension in the client searches for the text in each generated segment in the original HTML page. Before the beginning of each text segment, it injects a <h2> header containing the header text generated by OpenAI and an id with a header counter to identify the internal links;
(6) The browser extension injects a list of links in the beginning of the page with an internal link to each of the generated headers.

The source code has been made available online at https://github.com/raphaelwb/concept_proof_segmenter. The browser extension requires a ChatGPT token to be inserted at the indicated point in the code.

Figure 4 shows an example of an excerpt of a web page in Portuguese before processing by the browser extension. Figure 5 shows the exact text without processing in English. Figure 6 shows the processed text in Portuguese after using the browser extension, with the list of internal links to the headers and the first generated header. Figure 7 shows the exact text after processing translated into English.

5.3 Methods for User Evaluation of the Chrome Browser Extension Coupled with OpenAI

The evaluation of the Google Chrome browser extension implemented in the second development cycle followed a process similar to the evaluation conducted in the prototype used in the first development cycle, described in Section 4.4.

The tasks used the same four texts with the same three questions for each one. However, the second evaluation cycle aimed at a qualitative evaluation to obtain feedback from participants about the use of the browser extension. This evaluation cycle did not compare the use of the texts

Usar uma máscara PFF3 de alto grau pode proteger quase inteiramente os profissionais de saúde da covid, constatou a pesquisa.

A qualidade das máscaras faciais que os profissionais de saúde usam faz uma grande diferença no risco de infecção por coronavírus, segundo um estudo conduzido em um hospital em Cambridge, no Reino Unido.

Usar uma máscara de alto grau conhecida como PFF3 pode fornecer até 100% de proteção.

Já as máscaras cirúrgicas comuns apresentam um risco muito maior de se pegar o vírus, segundo a pesquisa.

Entidades de classe vem fazendo campanha para que trabalhadores recebam equipamentos melhores de proteção individual (EPI).

Os dados foram coletados durante um programa de testes da entidade NHS Foundation Trust, ligada ao sistema de saúde britânico, o NHS (equivalente ao SUS no Brasil).

Os resultados foram publicados em um artigo que ainda não foi revisado por pares.

Durante a maior parte do ano passado, o hospital seguiu a orientação do governo que especifica que os profissionais de saúde devem usar máscaras cirúrgicas, exceto em algumas situações limitadas.

Embora resistentes a fluidos, essas máscaras são relativamente frágeis e frouxas e não têm o objetivo de filtrar aerossóis infecciosos — minúsculas partículas de vírus que podem permanecer no ar e agora são amplamente tidas como fonte de infecção por coronavírus.

O estudo descobriu que a equipe que cuida de pacientes em alas "vermelhas" (onde há pacientes com covid) enfrentava um risco até 47 vezes maior do que aqueles em alas "verdes" (sem pacientes com covid).

Fig. 4. Original text from Text 2 without processing (in Portuguese).

Usar uma máscara PFF3 de alto grau pode proteger quase inteiramente os profissionais de saúde da covid, constatou a pesquisa.

A qualidade das máscaras faciais que os profissionais de saúde usam faz uma grande diferença no risco de infecção por coronavírus, segundo um estudo conduzido em um hospital em Cambridge, no Reino Unido.

Usar uma máscara de alto grau conhecida como PFF3 pode fornecer até 100% de proteção.

Já as máscaras cirúrgicas comuns apresentam um risco muito maior de se pegar o vírus, segundo a pesquisa.

Entidades de classe vem fazendo campanha para que trabalhadores recebam equipamentos melhores de proteção individual (EPI).

Os dados foram coletados durante um programa de testes da entidade NHS Foundation Trust, ligada ao sistema de saúde britânico, o NHS (equivalente ao SUS no Brasil).

Os resultados foram publicados em um artigo que ainda não foi revisado por pares.

Durante a maior parte do ano passado, o hospital seguiu a orientação do governo que especifica que os profissionais de saúde devem usar máscaras cirúrgicas, exceto em algumas situações limitadas.

Embora resistentes a fluidos, essas máscaras são relativamente frágeis e frouxas e não têm o objetivo de filtrar aerossóis infecciosos — minúsculas partículas de vírus que podem permanecer no ar e agora são amplamente tidas como fonte de infecção por coronavírus.

O estudo descobriu que a equipe que cuida de pacientes em alas "vermelhas" (onde há pacientes com covid) enfrentava um risco até 47 vezes maior do que aqueles em alas "verdes" (sem pacientes com covid).

Fig. 5. Original text from Text 2 without processing (in English).

Links para seções da página
Estudo Revela que o Risco de Infecção por Coronavírus Varia Muito de Acordo com a Máscara Usada.
Dados Coletados durante Programa de Testes da NHS Foundation Trust para o NHS (SUS) .
O Hospital de Cambridge Aumentou a Proteção em Meio à Segunda Onda da Pandemia de Coronavírus
Durante a maior parte do ano passado, o hospital seguiu a orientação do governo que
Ferris Revela que Adoção das Máscaras PFF3 Foi o Que Faltava Tentar Fazer
Estudo Conclui que Máscaras PFF3 Protegem Profissionais de Saúde Contra Covid.
Resultados Surpreendentes do Estudo sobre Máscaras PFF3 e Coronavírus.
Hospital de Cambridge entre 17 no Reino Unido Atualizando Máscaras Contra Política Nacional O Hospital de Cambridge está entre 17 hospitais em todo o Reino Unido que decidiram atualizar

Usar uma máscara PFF3 de alto grau pode proteger quase inteiramente os profissionais de saúde da covid, constatou a pesquisa.

A qualidade das máscaras faciais que os profissionais de saúde usam faz uma grande diferença no risco de infecção por coronavírus, segundo um estudo conduzido em um hospital em Cambridge, no Reino Unido.

Usar uma máscara de alto grau conhecida como PFF3 pode fornecer até 100% de proteção.

Já as máscaras cirúrgicas comuns apresentam um risco muito maior de se pegar o vírus, segundo a pesquisa.

Entidades de classe vem fazendo campanha para que trabalhadores recebam equipamentos melhores de proteção individual (EPI).

Dados Coletados durante Programa de Testes da NHS Foundation Trust para o NHS (SUS) .

Os dados foram coletados durante um programa de testes da entidade NHS Foundation Trust, ligada ao

Fig. 6. Processed text from Text 2 (in Portuguese).

with and without the extension, as in the first cycle. All participants were allowed to use the extension in all tasks.

Unlike the evaluation in cycle 1, we did not aim to measure workload and differences between using the browser extension and not using it. The main goal of this evaluation cycle was to gather qualitative feedback concerning the experience of using the browser extension and the automatically generated headers and links.

Due to using ChatGPT, the headers generated were different each time the browser extension was activated. This feature made it more difficult to compare the experiences that different users had when using the extension.

Evaluations also took place remotely. As the evaluations required that users install the browser extension remotely, this evaluation cycle required further experience with computers from participants than the evaluation in the first cycle. For this reason, the evaluation could not be conducted with the same user group as the previous evaluation.

5.4 Participants

In the evaluation of the browser extension developed in the second cycle, all participants were experienced screen-reader users with more than ten years of experience. Five participants used NVDA as their primary browser, one used JAWS, and one used VoiceOver.

Participants were recruited via snowballing from the researcher's social network. Table 7 summarises the main demographic data of participants who participated in the second user evaluation cycle.

Links to sections of the page
Study Reveals that the Risk of Coronavirus Infection Varies Greatly Depending on the Mask Used.
Data Collected During NHS Foundation Trust Testing Program for the NHS (SUS).
Cambridge Hospital Increased Protection Amid Second Wave of Coronavirus Pandemic For most of last year, the hospital followed government guidance Ferris Reveals
Adoption of PFF3 Masks Was What Was Left Trying to Make
Study Concludes that PFF3 Masks Protect Healthcare Professionals Against Covid.
Surprising Results of the Study on PFF3 Masks and Coronavirus.
Cambridge Hospital among 17 in UK Updating Masks Against National Policy Cambridge Hospital is among 17 hospitals across the UK that have decided to upgrade

Wearing a high-grade PFF3 mask can almost entirely protect healthcare workers from Covid, research finds.

The quality of the face masks that healthcare workers wear makes a big difference to their risk of coronavirus infection, according to a study conducted at a hospital in Cambridge, UK.

Using a high-grade mask known as PFF3 can provide up to 100% protection.

Common surgical masks, on the other hand, present a much higher risk of catching the virus, according to the research.

Trade associations have been campaigning for workers to receive better personal protective equipment (PPE).

Data Collected During NHS Foundation Trust Testing Program for the NHS (SUS).

The data was collected during a testing program by the NHS Foundation Trust, linked to the British

Fig. 7. Processed text from Text 2 (in Portuguese).

Table 7. Participants' Demographic Data—User Evaluation of the Browser Extension in the Second Cycle

Code	Age	Gender	Sight condition	Education level
P1-C2	30	F	Total blindness	Higher Education
P2-C2	28	M	Total blindness	Higher education
P3-C2	54	F	Total blindness	MSc degree
P4-C2	50	M	Total blindness	Post-graduate certificate
P5-C2	55	M	Total blindness	Higher education
P6-C2	30	M	Total blindness	PhD degree
P7-C2	68	M	Residual vision (dark/light)	Post-graduate certificate

5.5 Procedures

The tests involved installing the browser extension and performing the tasks independently. Unfortunately, the browser extension could only be installed in developer mode on Google Chrome. For this reason, the installation procedure was more challenging than expected. Participants had a synchronous session with one of the researchers to go through the installation process. Before the installation process, participants received the Informed Consent via a text document. The informed consent could also be read aloud if requested. After installing the extension, participants received an e-mail containing information about the text to be read using it, the questions they were trying to answer, and follow-up questions about their impressions of the extension.

The four texts with three questions each were the same as those reported in Section 4.5.
The following follow-up questions were posed to participants:

(1) Do you think the system made it easier to find information in the texts?
(2) Did the headers help to identify the content you needed to answer the questions?
(3) Did you have any difficulty when using the system?
(4) How do you think the system could be improved to make navigation easier?
(5) Was it easy to understand how to use the headers and internal links?

5.6 Results from User Evaluation in the Second Cycle

Six out of seven participants were able to complete all tasks successfully. One of the participants used VoiceOver with Safari as his primary browser. For this reason, he had more challenges in using Google Chrome.

Three out of the seven participants did not use navigation by headers and preferred to use the internal links to explore the content in the texts.

We conducted a content analysis on the feedback provided by the participants and identified the main trends in their perception of using the browser extension and the NLP algorithm.

5.6.1 The Design of Easy-to-Use Shortcuts. Some participants pointed out that the keyboard shortcut CTRL+SHIFT+Y was hard to use. They pointed out that using fewer keys and having a key closer to the edge of the keyboard would help them identify the extension's shortcut and avoid errors.

Participant P2-C2 pointed out that:

> "*As I am holding the CTRL and the SHIFT keys, it's harder to reach for the letter Y in the middle of the keyboard.*"

5.6.2 The Need for Feedback for Starting and Finishing Processing. Most participants pointed out that the processing was slow, and they did not have any feedback that the processing had started after they pressed the keyboard shortcut and when processing finished.

Participant P4-C2 stated:

> "*I missed a response when the feature is enabled. A pop-up could appear with the message that the screen reader software could read and provide us with this feedback.*"

5.6.3 Orientation and Sign-Posting in Navigation. Participants pointed out that they missed having short explanations of the navigation aids provided in the tool. They expected that the tool would briefly explain the internal links at the page's beginning. The explanation should state that the links were automatically generated and how many are listed so participants can mentally organize the number of links provided.

Participant P6-R2 pointed out:

> "*When I was using the TAB key to navigate in the document, I had to memorize the title of the first link so I could identify when I was returning to the beginning of the page. If they were numbered, it would be easier to figure out where I am.*"

5.6.4 Benefits from "Slicing" the Text. As in the evaluation in the first cycle, many participants pointed out that having automatically generated headings helped them organize their mental model of the page and spend less time reading through the text.

In many situations, participants also complained about the quality of the generated headers. Even in those situations, participants mentioned that they still could use the system to help identify different parts of the text.

Participant P1-C2 stated:

> "*As the system transforms (the text) into a link, the textual units, it is easier to find what you are looking for by just clicking in the paragraph that interests the user. Generally, sighted people do photographic and dynamic readings of large texts, memorizing the paragraph they need and thus, they can return to it more easily and quickly. What I could observe of the system is that if a person remembers the part of the paragraph they are looking for, You can also do the same dynamic reading.*"

Participant P3-C2 stated:

> "*A text without markings makes it difficult because you don't know where you will find the information to answer the question; you'll have to do an initial reading and then use search to find keywords.*"

5.6.5 Low Quality and Repeated Headers. ChatGPT did not consistently generate the same headers for all participants as they attempted to do their tasks. For many participants, there were situations in which two excerpts of the exact text had the same header generated by ChatGPT.

Participant P2-C2 stated:

> "The result was completely counterproductive, largely due to the fact that the tool generated information that was not very useful and disconnected in accordance with the communication objective of the content. In the text, New Coronavirus Infects and Replicates in Salivary Gland Cells, the links and titles created were as follows, in the order in which they were presented:
> 1. Discovery.
> 2. "New Study Reveals First Report of Respiratory Virus that Infects and Replicates in the Salivary Glands, Possibly Explaining the Infectivity of Sars.
> 3. Can Sars-CoV-2 Replicate in the Salivary Glands of Patients with Covid-19?
> 4. Virus Detection Using Molecular, Immunohistochemical and Electron Microscopy Analysis: Surprising Results!
> 5. New Study Will Investigate Whether the Mouth is a Direct Entry Point for Sars-CoV-2. Elderly people and ACE2 receptors in the mouth.
> These results do not resemble or come close to accessible content written by humans."

5.6.6 Integration with Screen Reader Features. Several participants pointed out that integrating the browser extension with existing features in screen readers was positive. Participants mentioned that they could use links or headings shortcuts from their screen reader after using the keyboard shortcut without learning extra features.

Participant P1-C2 stated:

> "*In addition to the Ctrl Shift Y shortcut, I used K and Shift K to navigate through the headings in the form of links.*"

Participant P5-C2 stated:

> "*After activating the plugin, the screen reader and browser commands are used for use.*"

5.6.7 Need for Multi-Level Heading Structure. Participants who used headings more proficiently complained about having only level-2 headings generated by the browser extension. Participants

said they would expect the system to make the main title clear and generate lower-level headings, which the system still needs to be able to do.

5.6.8 Avoiding Long Headers with Difficult Words. Participants complained that some of the headers generated by ChatGPT were too long and contained complex and redundant words that could make understanding more challenging.

5.6.9 Difficulty with Slow Processing. Participants pointed out that the long time it took to process the requests discouraged them from continuing to use the system.

Participant P4-C2 stated:

> "*I only noticed a slowness in converting the text without markup for the text with links and headings.*"

5.6.10 Limitations of Automatically Generated Content. Some participants acknowledged that the tool was limited in generating content without human revision. One participant alerted that he did not recommend using the tool instead of humans to generate content.

Participant P2-C2 stated:

> "*… this type of automated tool, in my view, has little to add without human review, as was possible to observe with the accessibility tests. Digital accessibility unfortunately cannot be solved with specific tools. The quality of what is generated, for now and especially in this case, falls short of what we would find with basic training for content creators, developers and designers.*"

5.6.11 Activating and Deactivating the Transformation. Participants complained that the system did not offer a feature to reverse the content to the previous state before processing.

Participant P4-C2 stated:

> "*I think the shortcut (control, shift and y) could be used to enable and disable the feature.*"

5.6.12 Inconstant Generation of Headers. Some participants complained about having different headings generated by ChatGPT. When participants reloaded the page, it was unclear to them that the page would have different headers and that they would have to rethink their strategies when navigating the system.

Participant P5-C2 stated:

> "*The plugin, …, does not have uniform behavior, that is, each time <Ctrl+shift+y> is activated, it will make a mark with a different title. The success or failure in searching for answers through links/headers will depend on the behavior of artificial intelligence."*

6 DISCUSSION

This project aimed to implement and evaluate a tool to automatically generate headers in long texts in Portuguese, which could be adapted as a plugin for screen readers to assist visually impaired users in information search tasks. For this, the project encompassed two cycles: (1) we developed an algorithm that automatically segments an input text in topic segments and generates headers based on keywords of each block. The algorithm was evaluated through a user study with 8 participants, all screen reader users, and (2) from the results from the first cycle, we conducted a cocreation

process of a browser extension with two blind experts and replaced the BERT-based labelling system with the OpenAI API for generating headers and internal links.

6.1 Analysis of the Algorithms—First Cycle

In this section, we discuss how our algorithms performed, comparing with previous studies and the implications of our results. In this project we implemented a topic segmentation algorithm adapting C99 to use BERT, and a topic labelling algorithm based on keywords. In conjunction, these algorithms enabled us to automatically generate headers.

We experimented with BERT in the topic segmenting algorithm to observe how this state-of-the-art model would improve an already established algorithm, C99. This decision was inspired by [40], which used Word2Vec and LSA to calculate sentence similarities. We first experimented using different layers of a BERT model to calculate cosine similarities, but the performance was slow. Speed was a vital aspect of this algorithm since screen reader users would have to wait for the process to finish to have the headers available. We then experimented with SBERT, which uses the GPU in the similarity calculation process, speeding up the process.

The results indicate that our algorithm did not improve the C99 algorithm. It had a higher error rate on detecting topic boundaries in most cases, which is worse. The only scenario where our algorithm yielded a better error rate than C99 was with the longest documents subset (9–11). This is a positive indicator since our project aimed to segment long texts for visually impaired users. Still, the time it takes to finish the segmentation process would heavily affect how users perceive this tool.

Further research can be done to assess if the error rate would improve for even longer documents, but time is a considerable limitation for our application. It also might be fruitful to try fine-tuning a BERT model specifically for the topic segmenting task in Portuguese and compare if it improves results. Nevertheless, in order to do that, a corpus with long segmented documents in Portuguese language is necessary.

For the labelling algorithm, we initially experimented with Bhatia et al.'s approach [13], which utilized both Word2Vec and Doc2Vec trained on a Wikipedia dump to compare the top-10 words for each topic and relate them to article titles from Wikipedia. However, our results were highly unsatisfactory, as the headers often had no connection with the text's subject matter. We suppose this is because, instead of labelling a document, we tried using this algorithm to label text segments, which were much smaller and lacked enough information. Another problem we had with this algorithm is that even when titles were related to the topic, they were still inadequate, limiting the meaning too much or making it too broad. These challenges suggested that even with improvements to the algorithm, it would still be inadequate for our task.

Even though machine learning algorithms can achieve impressive results, it is still necessary to have a corpus big enough to be correctly trained. Furthermore, this corpus needs to be task-dependent. In the case of our problem, a corpus with segmented and labelled texts for Portuguese would be needed. Even if we found it, when results are unsatisfactory, trying to pinpoint what influenced the result proved to be hard when we tried the Wikipedia-based label algorithm. The time necessary to train a new model also needs to be taken into account. For these reasons, we decided to focus the machine learning process on the segmenting algorithm and use a statistical algorithm for the labelling step.

6.2 Implications from the User Study in the First Cycle

Although the number of participants in the user study was insufficient to determine whether the proposed tool positively impacted participants, we can see an indication that cognitive load and

time measurements did decrease in texts with headers. Not only that, but users also reported they felt it was easier to complete tasks with automatically generated headers.

One interesting finding is that text segmentation appeared to be more helpful for the users than the headers themselves. Half of the participants reported not using the headers when completing the tasks, but even they agreed that having headers separating the text by topic helped navigate, locate themselves, and refind information in a second read. According to participants, this fact made the tasks with headers less exhausting, indicating how helpful navigation tools inside texts can be. Participants considering segmented texts easier to navigate is in line with what Tombaugh et al. [56] observed, that smaller fragments of texts–in our case topic segments–help locate portions of previously read texts by providing spatial cues, which aids in recollecting information.

Participant P5-C1 detailed why navigation tools are important, reading long texts in their entirety would take too much time, and be very exhausting when using a screen reader. He then described strategies applied when reviewing texts. One of such strategies is reading the start of paragraphs when looking for information because it often summarizes its subject, though this is not always the case. Another strategy for reading texts 70 pages long is sampling segments: reading the first, second paragraph, then the sixth, tenth, jumping through intervals to get a general sense of meaning. He also expressed discontent for screen readers not having the option to navigate by sentence. He either reads from the start of a paragraph or at the start of each line. However, he cannot choose a sentence in the middle of a paragraph.

Participant P7-C1 also reported a different strategy when reading long texts. He called it a *dynamic reading*: speeding up the screen reader to read the text fast without paying much attention and picking up keywords or concepts. Then, he searches for the keywords to read, paying more attention. This resembles the *Discovery* strategy pointed out by Power et al. [42], where the user gains an overview of a web-page. However, in this case, he applied it specifically for the text to locate topics of interest.

As observed by [16], everyday strategies to skip content included using header navigation shortcuts and keyword search; both were used for a second read. However, unlike what they observed, participants used sequential navigation as a first resort, skipping repeated content (such as the research title on every page). This corroborates to the difference we observed when comparing research about screen reader usage worldwide [65] and in Brazil [24]. Brazilians tend to read through the page much more than the rest of the world, where heading navigation is the norm.

In three instances, we observed what Power et al. [42] called *Reset* strategy. When a user abandons what they were doing, they start again at a safe point to work around an error. This situation happened twice when the screen reader lost focus on the web page and got stuck in the navigation bar area. The user could not figure out how to return to the page, so he restarted the browser. In another instance, Windows Update popped up, confusing the user, who decided to close every window and start again.

Although some users reported disliking the way words were presented, or even the choice of words to be used as a label, most users stated being able to infer the subject of topics by reading the label. We acknowledge there is still room for improvement in the labels, such as using syntax trees or language models when choosing words, but it was interesting to see this positive result. Participant P8-C1 indicated why the labels were intuitive. Despite being just a set of words, blind users often use keywords to find what they are looking for on a web page. Participant P2-C1 even reported feeling like he was "running his eyes through the text", our initial inspiration for this project, recreating text skimming for visually impaired people.

6.3 Implications from the Second Cycle of Development and Evaluation

The second development cycle provided essential insights from the requirements elicitation process with the participation of two blind experts as co-authors of this study and the evaluations

with seven participants. The two experts provided critical insights into how to interpret the user evaluation data from the first cycle and how to implement improvements, and the evaluation of the browser extension provided valuable insights into how the extension was used in a situation closer to a real-world scenario.

The use of OpenAI and ChatGPT provided essential contributions to improving the generation of labels. As indicated by previous studies [17, 33], advanced NLP tools in the OpenAI model have vital potential to improve assistive technologies.

The headers generated during the tests provided a better understanding of the topics in the analysis of the experts in different fields involved in the study. However, it is still necessary to perform more extended studies to analyze the types of headers generated on different occasions with a broader range of texts and whether there are significant changes in the headers generated in different situations. As the content generated by ChatGPT is not constant in different interactions, not all participants experienced the same quality of text. For many participants, the output generated by ChatGPT was still not easy to understand by them.

Some participants had experience in the area of digital content design. These participants were very critical of the prospect of having automated AI-based tools as a substitute for human-generated content. They pointed out several limitations in different stances of content generation and how some outputs may seriously compromise the understanding, with issues such as difficult-to-understand words and repeated headers.

For future developments, the results from the study show possible avenues for research into how the use of LLMs could be adjusted to provide better results when coupled with assistive technologies. However, the results clearly show that such tools are only limited to alleviating the problems that would need human intervention for effective solving.

6.4 Limitations

The current implementation of the browser extension developed in the second cycle had some limitations in its features. The current version of the algorithm only analyzes textual information on web pages. Information in images, tables and other interactive elements, such as forms and buttons, isstill need to be analyzed. Future versions could advance by devising strategies to integrate such content into the analysis. Additionally, the tool has limitations as it still depends on server-side processing and requires a key from OpenAI. However, future versions will allow users to provide their own key to enable the processing of requests.

The COVID pandemic caused another limitation in the first cycle of development. User tests had to be conducted remotely, forcing us to recruit participants with computer and screen reader experience. This fact influences the results since users with basic computer knowledge or beginner screen-reader users might encounter other problems and give further feedback. The impact of the experience was reinforced by the two blind experts involved in the second development cycle. They reinforced the importance of catering for screen-reader users with different experience levels. This result led, for example, to implementing features to aid navigation by internal links and not only headings.

Even though participants in the evaluation in the first cycle were experienced screen reader users, the tests were not without problems. When doing tests, the main issue was the screen reader occasionally changing the focus from the web page to the address bar. This problem can generally happen when reading web pages not specific to our test.

The number of participants in the first evaluation cycle was too small to cover all possible combinations between texts and scenarios (texts with and without automatically generated headers. However, scenarios were alternated between participants to lower the influence on the results.

Due to time restrictions for an online test–the total time for each test ranged from 43 minutes up to 2 hours–, we had to compromise and choose texts smaller than our initial target to avoid fatigue on participants. Future works in a laboratory, done in more than a day, could explore longer texts.

The second evaluation cycle was limited to seven participants, all of whom had to be experienced screen-reader users. As the first prototype had to be installed in developer mode on Google Chrome, it would be considerably more challenging to evaluate with participants with more limited experience. Future studies will involve the evaluation including a broader range of participants with varied experience levels.

Our results from the user evaluation in the first cycle showed that, although users saw the benefit of using the proposed tool, the header labels still need to be improved. They needed improvements in how they were presented and in the choice of words. There is no document-wide awareness in the present state, as each topic is labelled separately. This limitation causes repetitive words throughout the document to appear in many headers, not adding value in differentiating one segment from another.

The use of OpenAI provided improvements in the quality of the headers generated. However, many participants still pointed out that the quality of the generated headings still had problems. The nature of the requests to ChatGPT also makes the generated headings different each time the algorithm is executed. Different participants experienced different generated headings with different quality. For this reason, conducting a comparative study of the headings was impossible, as in the first evaluation cycle.

The present study has not evaluated the long-term acceptance of the tool and its interplay with broader accessibility strategies. It was clear from participants' feedback in the second evaluation cycle that the use of AI tools is not a replacement for appropriate accessible development. The tool can alleviate issues caused by poorly designed websites, but human-generated content is still essential. Future studies could explore the long-term impact of AI-based tools, such as the browser extension developed to overcome accessibility problems.

7 CONCLUSION AND FUTURE WORK

This research aimed to explore the feasibility of using automatic header generation to enhance the experience of screen reader users while browsing web content for information. To accomplish this, we conducted the study in two cycles. In the first cycle, we created an algorithm that used text segmentation and topic labelling techniques to generate headers automatically. Additionally, we examined the effectiveness of advanced word embedding models, such as C99 with BERT, in improving the algorithm's performance. The outcomes of our investigation demonstrated that C99 with BERT did not outperform prior findings in most cases.

The user study with eight participants in the first cycle indicated that the segmentation process was good enough for users to benefit from using the tool. Participants had positive reactions when interacting with texts processed by our proposed algorithm. They reported that the topic segment separation helped locate information and navigate the text.

Most participants also reported being able to infer the subject of a text segment by its label and that the headers helped when recollecting information in a second read. Other participants argued that their choice of words and presentation could be improved. Even so, all users were eager to have a similar tool available for their screen readers to have more navigation options.

To alleviate the problems pointed out by users in the labels of textual segments, we presented a second cycle of development and our first experience using Open AI and ChatGPT to generate labels from a segmented text. ChatGPT can generate topics in a text based on analysing the words, phrases, and concepts contained within it. From this, we believe that ChatGPT can determine which words are most relevant to each topic segment in a text and select them as topic labels.

Still, there is room for improvement. Future work may explore different segmentation and keyword generation techniques, as these are rapidly evolving research fields. For example, recent studies have shown the use of LLM suitable for keyword generation, such as the *Galactia* model, which was trained on a large volume of text and demonstrated excellent performance when compared to solutions that did not use LLM [34]. In another approach, Lee et al. [8] introduced an approach for effectively segmenting and classifying text while relying on latent topics. The end-to-end method builds on a neural topic embedding, trained using Wikipedia headings to optimize a bidirectional LSTM classifier. The authors propose that this method holds promise for extracting and retrieving relevant content from unannotated collections, such as medical research articles or technical papers.

The proposed tool was too slow to be implemented client-side, and word embedding models proved to have limitations for this problem. A corpus of segmented texts in Portuguese would be beneficial for training models or fine-tuning BERT specifically for this task. In order to be applied as an actual plugin, the tool also needs to be faster since waiting times would impact usability. Besides that, we also plan to explore how ChatGPT can be useful for people with disabilities, especially when searching the web for information of interest.

The COVID pandemic also limited our options for the user study in the first cycle. The limitations in installing the browser extension in developer mode also limited the second evaluation cycle to more experienced participants. Future work could involve participants of different ranges of experience with screen readers or computers. Testing with longer documents is also important. Future studies will cover the use of the current browser extension with different types of texts and different types of texts generated by ChatGPT.

The results from the study raised several issues that can open research avenues into the use of generative artificial intelligence and NLP in assistive technologies. Participants in both development cycles pointed out that the automatically generated headers served as a means to alleviate problems in inaccessible pages. However, the quality of the generated content still needs to be improved to replace content generated by humans.

After comparing the two development cycles, the study also showed the need to pursue research into different types of language models to generate input for assistive technologies. The current model depends on resources requiring high processing power running on centralized servers. Investigating ways to run the models with lower computation costs that could run on the client would be essential.

APPENDICES

A TEXT 1 USED IN THE EVALUATIONS

A.1 Text 1 in Portuguese

Novo coronavírus infecta e se replica em células das glândulas salivares

Pesquisadores da Faculdade de Medicina da Universidade de São Paulo (FMUSP) constataram que o coronavírus infecta e se replica em células das glândulas salivares.

Por meio de análises de amostras de três tipos de glândulas salivares, obtidas durante um procedimento de autópsia minimamente invasiva em pacientes que morreram em decorrência de complicações da Covid-19 no Hospital das Clínicas da FMUSP, eles verificaram que esses tecidos especializados na produção e secreção de saliva são reservatórios para o novo coronavírus. Os resultados do estudo, apoiado pela Fapesp, foram publicados no Journal of Pathology.

As descobertas contribuem para explicar por que o novo coronavírus é encontrado em grandes quantidades na saliva, o que viabilizou a realização de testes para diagnósticos da Covid-19 a partir do fluido, sublinham os autores do trabalho.

"É o primeiro relato de vírus respiratório capaz de infectar e se replicar nas glândulas salivares. Até então, acreditava-se que apenas vírus causadores de doenças com prevalência muito alta, como o da herpes, usavam as glândulas salivares como reservatório. Isso pode ajudar a explicar por que o Sars-CoV-2 é tão infeccioso", diz à Agência Fapesp Bruno Fernandes Matuck, doutorando na Faculdade de Odontologia da USP e primeiro autor do estudo.

Os pesquisadores já tinham demonstrado, em estudo anterior, a presença de RNA do Sars-CoV-2 no tecido periodontal de pacientes que morreram em decorrência da Covid-19.

Em razão da alta infecciosidade do Sars-CoV-2 quando comparado a outros vírus respiratórios, eles levantaram a hipótese de que o novo coronavírus poderia infectar e se replicar em células das glândulas salivares e, dessa forma, surgir na saliva sem ter contato com secreções nasais e pulmonares.

Isso porque estudos internacionais anteriores mostraram que o ducto salivar apresenta o receptor ACE2, com o qual a proteína spike do Sars-CoV-2 se liga para infectar as células. Mais recentemente, outros grupos de cientistas relataram ter observado em estudos feitos com animais que, além da ACE2, receptores como a serina protease transmembranar 2 (TMPRSS) e a furina, presentes nos tecidos das glândulas salivares, são alvos do Sars-CoV-2.

A fim de testar essa hipótese em humanos, foram feitas biópsias guiadas por ultrassom em 24 pacientes que morreram em decorrência da Covid-19, com idade média de 53 anos, para extração de amostras de tecidos das glândulas parótida, submandibular e menores.

As amostras dos tecidos foram submetidas a análises moleculares (RT-PCR) para identificação da presença do vírus. Os resultados indicaram a presença do vírus nos tecidos em mais de dois terços das amostras.

Já por meio de marcações imuno-histoquímicas – em que é colocado um corante em uma molécula que se gruda no vírus e nos receptores –, foi possível observar a presença do vírus in situ, no interior dos tecidos. E, por meio de microscopia eletrônica, foi detectada não só a presença, mas também o vírus se replicando nas células e identificado o tipo de organela que ele utiliza para essa finalidade.

"Observamos vários vírus aglomerados nas células das glândulas salivares – um indicativo de que estão se replicando em seu interior. Não estavam presentes nessas células passivamente", afirma Matuck. Os pesquisadores pretendem avaliar, agora, se a boca pode ser uma porta de entrada direta do Sars- CoV-2, uma vez que os receptores ACE2 e o TMPRSS são encontrados em vários locais da cavidade, como em tecidos da gengiva e da mucosa bucal. Além disso, a boca tem uma área de contato maior do que a cavidade nasal, apontada como a principal porta de entrada do vírus.

"Por meio de uma parceria com pesquisadores da Universidade da Carolina do Norte, dos Estados Unidos, pretendemos mapear a distribuição desses receptores na boca e quantificar as replicações virais em tecidos bucais", diz Luiz Fernando Ferraz da Silva, professor da FMUSP e coordenador do projeto. "Pode ser que a boca seja um meio viável para entrada direta do vírus", estima Matuck.

Outra ideia é verificar se idosos possuem mais receptores ACE2 na boca em comparação com pessoas mais jovens, uma vez que têm uma diminuição do fluxo salivar. A despeito disso, os pesquisadores encontraram mesmo em pacientes idosos, que têm menos tecidos salivares, uma alta carga viral. "Esses pacientes quase não tinham tecido salivar, era quase tudo tecido gorduroso. Mas, mesmo assim, ainda apresentavam uma carga viral relativamente alta", relata Matuck.

A.2 Text 1 in English

New coronavirus infects and replicates in salivary gland cells

Researchers from the Faculty of Medicine of the University of São Paulo (FMUSP) found that the coronavirus infects and replicates in salivary gland cells.

By analyzing samples of three types of salivary glands, obtained during a minimally invasive autopsy procedure in patients who died as a result of complications from Covid-19 at the FMUSP Hospital das Clínicas, they verified that these tissues specialized in production and secretion of saliva are reservoirs for the new coronavirus. The results of the study, supported by Fapesp, were published in the Journal of Pathology.

The findings contribute to explaining why the new coronavirus is found in large quantities in saliva, which made it possible to carry out tests for diagnosing Covid-19 using the fluid, emphasize the authors of the work.

"It is the first report of a respiratory virus capable of infecting and replicating in the salivary glands. Until then, it was believed that only viruses that caused very high prevalence diseases, such as herpes, used the salivary glands as a reservoir. This could help explain why Sars-CoV-2 is so infectious", Bruno Fernandes Matuck, PhD student at the USP School of Dentistry and first author of the study, tells Agência Fapesp.

The researchers had already demonstrated, in a previous study, the presence of Sars-CoV-2 RNA in the periodontal tissue of patients who died as a result of Covid-19.

Due to the high infectivity of Sars-CoV-2 when compared to other respiratory viruses, they raised the hypothesis that the new coronavirus could infect and replicate in salivary gland cells and, thus, appear in saliva without coming into contact with secretions. nasal and pulmonary.

This is because previous international studies have shown that the salivary duct has the ACE2 receptor, with which the Sars-CoV-2 spike protein binds to infect cells. More recently, other groups of scientists reported having observed in animal studies that, in addition to ACE2, receptors such as transmembrane serine protease 2 (TMPRSS) and furin, present in salivary gland tissues, are targets of Sars-CoV-2.

In order to test this hypothesis in humans, ultrasound-guided biopsies were performed on 24 patients who died as a result of Covid-19, with an average age of 53 years, to extract tissue samples from the parotid, submandibular and minor glands.

Tissue samples were subjected to molecular analysis (RT-PCR) to identify the presence of the virus. The results indicated the presence of the virus in the tissues in more than two thirds of the samples.

Using immunohistochemical markings – in which a dye is placed on a molecule that sticks to the virus and receptors – it was possible to observe the presence of the virus in situ, inside the tissues. And, using electron microscopy, not only the presence, but also the virus replicating in cells was detected and the type of organelle it uses for this purpose was identified.

"We observed several viruses clustered in salivary gland cells – an indication that they are replicating inside them. They were not present in these cells passively", says Matuck. The researchers now intend to evaluate whether the mouth can be a direct entry point for Sars-CoV-2, since ACE2 and TMPRSS receptors are found in various locations in the cavity, such as in gum tissue and oral mucosa. Furthermore, the mouth has a larger contact area than the nasal cavity, considered the main entry point for the virus.

"Through a partnership with researchers from the University of North Carolina, in the United States, we intend to map the distribution of these receptors in the mouth and quantify viral replications in oral tissues", says Luiz Fernando Ferraz da Silva, professor at FMUSP and coordinator of the project. "It may be that the mouth is a viable means of direct entry of the virus", estimates Matuck.

Another idea is to check whether elderly people have more ACE2 receptors in their mouth compared to younger people, since they have a decrease in salivary flow. Despite this, researchers found even in elderly patients, who have less salivary tissue, a high viral load. "These patients had

almost no salivary tissue, it was almost all fatty tissue. But even so, they still had a relatively high viral load", reports Matuck.

A.3 Questions for Text 1

(1) The text talks about a place where the new coronavirus replicates. What is this location on the body?
(2) Is there a difference in the number of salivary glands between elderly and young people?
(3) In what percentage of samples was the virus found in the study?

B TEXT 2 USED IN THE EVALUATIONS

B.1 Text 2 in Portuguese

Usar uma máscara PFF3 de alto grau pode proteger quase inteiramente os profissionais de saúde da covid, constatou a pesquisa.

A qualidade das máscaras faciais que os profissionais de saúde usam faz uma grande diferença no risco de infecção por coronavírus, segundo um estudo conduzido em um hospital em Cambridge, no Reino Unido.

Usar uma máscara de alto grau conhecida como PFF3 pode fornecer até 100% de proteção.

Já as máscaras cirúrgicas comuns apresentam um risco muito maior de se pegar o vírus, segundo a pesquisa.

Entidades de classe vem fazendo campanha para que trabalhadores recebam equipamentos melhores de proteção individual (EPI).

Os dados foram coletados durante um programa de testes da entidade NHS Foundation Trust, ligada ao sistema de saúde britânico, o NHS (equivalente ao SUS no Brasil).

Os resultados foram publicados em um artigo que ainda não foi revisado por pares.

Durante a maior parte do ano passado, o hospital seguiu a orientação do governo que especifica que os profissionais de saúde devem usar máscaras cirúrgicas, exceto em algumas situações limitadas.

Embora resistentes a fluidos, essas máscaras são relativamente frágeis e frouxas e não têm o objetivo de filtrar aerossóis infecciosos — minúsculas partículas de vírus que podem permanecer no ar e agora são amplamente tidas como fonte de infecção por coronavírus.

O estudo descobriu que a equipe que cuida de pacientes em alas "vermelhas" (onde há pacientes com covid) enfrentava um risco até 47 vezes maior do que aqueles em alas "verdes" (sem pacientes com covid).

O pesquisador principal, Mark Ferris, especialista em saúde ocupacional do hospital, disse que a equipe estava se contaminando com covid, apesar de seguir todos os protocolos de controle de infecção.

Assim, quando a segunda onda da pandemia começou a acontecer em dezembro passado, os gerentes dos hospitais de Cambridge tomaram a decisão de aumentar a proteção nas alas vermelhas. "A única coisa que faltava tentar fazer ainda era adotar as máscaras PFF3, e foi isso que eles fizeram", disse Ferris.

As máscaras PFF3 são mais fechadas no rosto e projetadas especificamente para filtrar aerossóis. Nas semanas seguintes a essa mudança, a taxa de infecções entre os profissionais de saúde nas enfermarias vermelhas caiu drasticamente, atingindo o nível das alas verdes, onde não havia pacientes com covid.

O estudo conclui que "os casos atribuídos à exposição na ala caíram significativamente, com máscaras PFF3 fornecendo 31–100% de proteção (e provavelmente 100%) contra a infecção de pacientes com covid-19."

Todos os casos restantes provavelmente foram causados por disseminação na comunidade, e não no hospital.

O artigo afirma que as máscaras cirúrgicas resistentes a fluidos são "insuficientes" para proteger os profissionais de saúde.

Mike Weekes, da Cambridge University NHS Hospitals Foundation Trust, que trabalhou no estudo, disse que a pesquisa dá "algumas evidências do mundo real de que as máscaras PFF3 são realmente eficazes, e mais eficazes do que as máscaras cirúrgicas".

Ele acrescentou: "Claramente, é um estudo relativamente pequeno e, portanto, precisamos ver essas descobertas replicadas em outro lugar. Mas, dada a diferença nos resultados que vimos, como uma espécie de efeito do princípio de precaução, o que devemos pensar é adotar máscaras PFF3 para quem cuida de um paciente com coronavírus."

O hospital de Cambridge está entre 17 em todo o Reino Unido que decidiram atualizar suas máscaras, à revelia da política nacional.

O apelo para que as máscaras PFF3 sejam distribuídas de forma mais ampla é uma demanda de longa data de diversos órgãos de profissionais de saúde.

Em uma carta aberta ao Secretário de Saúde do Reino Unido, Sajid Javid, um grupo de consultores e médicos diz que o novo estudo fornece ainda mais evidências de por que a política precisa ser alterada.

"Isso tem implicações importantes para a proteção do trabalhador de saúde, já que o Reino Unido lida com o que se espera que seja uma 'onda final', além de tentar reduzir o enorme acúmulo de outros trabalhos enquanto enfrenta a inevitável doença e isolamento da equipe", escreveu o grupo.

Um porta-voz do Departamento de Saúde e Assistência Social disse que as orientações sobre os padrões de máscaras são atualizadas regularmente para refletir os mais recentes avanços científicos.

"A segurança do NHS (sistema de saúde britânico) e da equipe de assistência social sempre foi nossa prioridade e continuamos a trabalhar 24 horas por dia para entregar materiais para proteger aqueles na linha de frente. As evidências e os dados emergentes são monitorados e revisados continuamente e as orientações serão alteradas de acordo, caso seja apropriado."

B.2 Text 2 in English

Wearing a high-grade PFF3 mask can almost entirely protect healthcare workers from Covid, research finds.

The quality of the face masks that healthcare workers wear makes a big difference to their risk of coronavirus infection, according to a study conducted at a hospital in Cambridge, UK.

Using a high-grade mask known as PFF3 can provide up to 100% protection.

Common surgical masks, on the other hand, present a much higher risk of catching the virus, according to the research.

Trade associations have been campaigning for workers to receive better personal protective equipment (PPE).

The data was collected during a testing program by the NHS Foundation Trust, linked to the British healthcare system, the NHS (equivalent to SUS in Brazil).

The results were published in a paper that has not yet been peer-reviewed.

For most of the past year, the hospital has followed government guidance that specifies that healthcare workers should wear surgical masks except in some limited situations.

Although resistant to fluids, these masks are relatively fragile and loose and are not intended to filter infectious aerosols — tiny virus particles that can linger in the air and are now widely thought to be a source of coronavirus infection.

The study found that staff caring for patients on "red" wards (where there are Covid patients) faced up to 47 times greater risk than those on "green" wards (without Covid patients).

Lead researcher Mark Ferris, an occupational health specialist at the hospital, said staff were becoming infected with Covid despite following all infection control protocols.

So, when the second wave of the pandemic began to hit last December, Cambridge hospital managers made the decision to increase protection on red wards. "The only thing left to try to do was adopt PFF3 masks, and that's what they did," Ferris said.

PFF3 masks are tighter to the face and specifically designed to filter aerosols. In the weeks following this change, the rate of infections among healthcare workers in red wards fell sharply, reaching the level of green wards, where there were no Covid patients.

The study concludes that "cases attributed to exposure on the ward have dropped significantly, with PFF3 masks providing 31–100% (and likely 100%) protection against infection of Covid-19 patients."

All remaining cases were likely caused by community spread rather than hospital spread.

The article states that fluid-resistant surgical masks are "insufficient" to protect healthcare workers.

Mike Weekes, from Cambridge University NHS Hospitals Foundation Trust, who worked on the study, said the research gives "some real-world evidence that PFF3 masks are indeed effective, and more effective than surgical masks".

He added: "Clearly, it's a relatively small study and so we need to see these findings replicated elsewhere. But given the difference in results that we saw, as a kind of effect of the precautionary principle, what we should think about is adopting PFF3 masks for those caring for a coronavirus patient."

The Cambridge hospital is among 17 across the UK that have decided to update their masks, despite national policy.

The call for PFF3 masks to be distributed more widely has been a long-standing demand from several healthcare professional bodies.

In an open letter to UK Health Secretary Sajid Javid, a group of consultants and doctors say the new study provides even more evidence as to why the policy needs to be changed.

"This has important implications for healthcare worker protection as the UK deals with what is expected to be a 'final wave', as well as trying to reduce the huge backlog of other work whilst facing the inevitable illness and isolation of team," the group wrote.

A Department of Health and Social Care spokesperson said guidance on mask standards is regularly updated to reflect the latest scientific advances.

"The safety of NHS and social care staff has always been our priority and we continue to work 24 hours a day to deliver materials to protect those on the frontline. Emerging evidence and data is monitored and reviewed continually and guidance will be amended accordingly if appropriate."

B.3 Questions for Text 2

(1) Why do PFF3 masks perform better?
(2) What entity collected the survey data?
(3) What conclusion did the research reach regarding the use of masks?

C TEXT 3 USED IN THE EVALUATIONS

C.1 Text 3 in Portuguese

Você já sentiu como se todo mundo tivesse mais a agradecer do que você? Cheque o seu feed o Facebook ou do Instagram: seus amigos parecem jantar mais em restaurantes chiques, tirar mais férias exóticas e ter filhos mais bem-sucedidos. Eles têm até bichos de estimação mais fofos!

Pode ter certeza de que isso é uma ilusão, que está enraizada em uma propriedade das relações sociais conhecida como o paradoxo da amizade. Inicialmente formulado pelo sociólogo Scott Feld, o paradoxo afirma que "seus amigos são, em média, mais populares do que você". Essa propriedade se combina a outras peculiaridades das interações sociais para criar uma ilusão.

O que o paradoxo significa é: se eu te perguntasse quem são os seus amigos e então os conhecesse, de forma geral eu consideraria que eles têm mais conexões sociais do que você. É claro, se você é uma pessoa extremamente sociável, o paradoxo não se aplicará a você. Mas para a maioria de nós, é provável que ele se aplique.

Embora esse paradoxo possa ocorrer em qualquer relação social, isso acontece de forma desenfreada online. Um estudo concluiu que 98% dos usuários do Twitter seguem contas que têm mais seguidores do que eles.

Embora soe estranho, o paradoxo da amizade tem uma explicação matemática simples. O círculo pessoal de cada pessoa é diferente. A maioria de nós tem alguns amigos, e aí existem as pessoas bem conectadas, como David Rockefeller, antigo CEO do Chase Manhattan Bank, cuja agenda inclui mais de 100 mil pessoas.

Nas redes sociais, celebridades como Justin Bieber podem ter mais de 100 milhões de seguidores. É esse pequeno grupo de pessoas superconectadas—pessoas com muitos amigos, que são parte do seu círculo social—que aumentam a popularidade média dos seus amigos.

Esse é o truque matemático no centro do paradoxo da amizade. A popularidade extraordinária da pessoas como Justin Bieber não só desvirtua a popularidade média de amigos para qualquer um que esteja conectado a eles, mas mesmo que pessoas como ele sejam raras, elas também aparecem em um número extraordinário de círculos sociais.

E o paradoxo da amizade não é apenas uma mera curiosidade matemática. Ele tem aplicações úteis em prever tendências e monitorar doenças. Pesquisadores o utilizaram para prever temas em alta no Twitter semanas antes de eles acontecerem e para identificar surtos de gripe em suas fases iniciais e elaborar estratégias eficientes para gerenciar a doença.

Funciona assim: imagine, por exemplo, que você chega a uma vila africana com apenas cinco doses de vacina contra o Ebola. A melhor estratégia não é vacinar as primeiras cinco pessoas que você encontrar, mas sim perguntar a essas pessoa quem são os seus amigos e vacinar esses cinco amigos. Se você fizer isso, é mais provável que você escolha pessoas que têm círculos sociais mais amplos e assim poderiam infectar mais pessoas se ficassem doentes. Vacinar os amigos seria mais eficiente para conter a disseminação do Ebola do que imunizar pessoas aleatórias que podem estar na periferia de um círculo social e não conectadas a muitas outras pessoas.

Tem mais. Notavelmente, uma versão mais forte do paradoxo da amizade se aplica a muitas pessoas. A maior parte dos seus amigos têm mais amigos do que você. Pense nisso. Não estou mais falando sobre médias, onde um amigo excepcionalmente popular poderia desvirtuar a média de popularidade dos seus amigos.

O que isso significa é que a maioria dos seus amigos é mais bem conectada socialmente do que você. Vá em frente e experimente. Clique no nome de cada um dos seus amigos no Twitter e veja quantos seguidores eles têm e quantas contas eles seguem. Estou disposta a apostar que os números são maiores do que os seus.

Ainda mais estranho, esse paradoxo se aplica não só à popularidade, mas também a outras características, como o entusiasmo ao usar redes sociais, jantares em bons restaurantes, ou tirar férias em lugares exóticos. Como um exemplo concreto, veja com que frequência alguém posta atualizações no Twitter. É verdade que a maioria das pessoas que você segue posta mais atualizações do que você. Além disso, a maioria das pessoas que você segue recebe mais informações novas e diversas do que você. E a maioria das pessoas que você segue recebe mais informações virais que acabam se espalhando no seu feed mais do que você imagina.

Essa versão mais forte do paradoxo da amizade pode levar a uma ilusão de maioria, na qual uma característica que é rara em uma rede de forma geral parece comum em muitos círculos sociais. Imagine que poucas pessoas, no geral, são ruivas, mas ainda assim, para muitas pessoas, parece que a maioria dos seus amigos tem cabelo ruivo. Tudo o que precisamos para disseminar a ilusão de que o cabelo ruivo é comum é que alguns influenciadores superconectados sejam ruivos.

A ilusão da maioria pode explicar por que você pode achar que seus amigos parecem estar fazendo coisas mais empolgantes: pessoas que são mais conectadas socialmente influenciam desproporcionalmente o que vemos e aprendemos nas redes sociais. Isso ajuda a explicar por que adolescentes superestimam a prevalência de bebedeiras em campi de universidades e por que alguns temas parecem ser mais populares no Twitter do que realmente são.

A ilusão da maioria pode distorcer as suas percepções sobre a vida de outros. Pessoas mais bem conectadas também fazem coisas mais notáveis, como jantar em um restaurante premiado ou tirar férias em Bora Bora. Elas também são mais ativas em redes sociais e mais propensas a registrar a vida no Instagram, distorcendo as nossas percepções de quão comuns essas coisas realmente são. Um bom jeito de mitigar essa ilusão é parar de se comparar com amigos e ser grato pelo que você tem.

C.2 Text 3 in English

Have you ever felt like everyone else has more to be thankful for than you? Check your Facebook or Instagram feed: Your friends seem to dine at more fancy restaurants, take more exotic vacations, and have more successful children. They even have cuter pets!

You can be sure that this is an illusion, which is rooted in a property of social relations known as the friendship paradox. Initially formulated by sociologist Scott Feld, the paradox states that "your friends are, on average, more popular than you." This property combines with other peculiarities of social interactions to create an illusion.

What the paradox means is: if I asked you who your friends were and then met them, I would generally consider them to have more social connections than you. Of course, if you are an extremely sociable person, the paradox will not apply to you. But for most of us, it probably applies.

Although this paradox can occur in any social relationship, it happens rampant online. A study concluded that 98% of Twitter users follow accounts that have more followers than them.

Although it sounds strange, the friendship paradox has a simple mathematical explanation. Everyone's personal circle is different. Most of us have a few friends, and then there are the well-connected people, like David Rockefeller, former CEO of Chase Manhattan Bank, whose diary includes more than 100,000 people.

On social media, celebrities like Justin Bieber can have more than 100 million followers. It's this small group of super-connected people—people with lots of friends, who are part of your social circle—that increase your friends' average popularity.

This is the mathematical trick at the heart of the friendship paradox. The extraordinary popularity of people like Justin Bieber not only skews the average popularity of friends for anyone who is connected to them, but even though people like him are rare, they also appear in an extraordinary number of social circles.

And the friendship paradox is not just a mere mathematical curiosity. It has useful applications in predicting trends and monitoring diseases. Researchers have used it to predict trending topics on Twitter weeks before they happen, and to identify flu outbreaks in their early stages and devise effective strategies for managing the disease.

It works like this: imagine, for example, that you arrive in an African village with just five doses of the Ebola vaccine. The best strategy is not to vaccinate the first five people you meet, but rather to ask those people who their friends are and vaccinate those five friends. If you do this, you are

more likely to choose people who have wider social circles and so could infect more people if they got sick. Vaccinating friends would be more effective in curbing the spread of Ebola than immunizing random people who may be on the periphery of a social circle and not connected to many other people.

Have more. Notably, a stronger version of the friendship paradox applies to many people. Most of your friends have more friends than you. Think about it. I'm no longer talking about averages, where an exceptionally popular friend could distort the average popularity of your friends.

What this means is that most of your friends are better socially connected than you are. Go ahead and try it. Click on each of your friends' names on Twitter and see how many followers they have and how many accounts they follow. I'm willing to bet the numbers are higher than yours.

Even stranger, this paradox applies not only to popularity, but also to other characteristics, such as enthusiasm for using social media, dining in fine restaurants, or taking vacations in exotic places. As a concrete example, look at how often someone posts updates on Twitter. It's true that most people you follow post more updates than you do. Plus, most people you follow receive more new and diverse information than you do. And most of the people you follow receive more viral information that ends up spreading in your feed more than you realize.

This stronger version of the friendship paradox can lead to a majority illusion, in which a trait that is rare in a network at large appears common in many social circles. Imagine that few people in general are redheads, but still, for many people, it seems like most of their friends have red hair. All it takes to spread the illusion that red hair is common is for a few super-connected influencers to be redheads.

The majority delusion may explain why you might think your friends seem to be doing more exciting things: People who are more socially connected disproportionately influence what we see and learn on social media. This helps explain why teens overestimate the prevalence of binge drinking on college campuses and why some topics appear to be more popular on Twitter than they really are.

The illusion of the majority can distort your perceptions of the lives of others. Better-connected people also do more notable things, like dining at an award-winning restaurant or vacationing in Bora Bora. They are also more active on social media and more likely to record their lives on Instagram, distorting our perceptions of how common these things really are. A good way to mitigate this illusion is to stop comparing yourself to friends and be grateful for what you have.

C.3 Questions for Text 3

(1) Can more connected people influence more?
(2) How can vaccinating friends help prevent diseases?
(3) What is the perception of teenagers about parties on social media?

D TEXT 4 USED IN THE EVALUATIONS

D.1 Text 4 in Portuguese

Começam hoje as inscrições para a edição 2021 do Exame Nacional do Ensino Médio (Enem), abertas até o 14 de julho. As provas estão marcadas para os dias 21 e 28 de novembro, nesta que será a segunda edição do Enem desde o início da pandemia de COVID-19. Especialistas avaliam que não haverá grande mudanças no que se refere ao conteúdo e apostam na redução da tensão em relação à última edição, com estudantes agora mais acostumados ao ensino a distância e até mesmo já podendo acessar as aulas presenciais nos cursinhos preparatórios. Lembram, no entanto, que essa adaptação é desigual, principalmente para os pobres. Um calendário mais definido também

contribui para uma expectativa melhor, na visão de alunos que enfrentaram as incertezas na versão passada do Enem, realizada este ano depois de adiamentos.

Para se inscrever, os candidatos devem acessar a página do participante e escolher entre fazer a prova impressa ou digital. Este ano, o Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira (Inep), responsável pela realização do Enem, vai oferecer mais de 101 mil inscrições para a versão digital da prova. Apenas alunos que já concluíram o ensino médio ou vão concluir em 2021 podem optar por esse modelo. As provas impressa e digital serão feitas na mesma data e, portanto, terão as mesmas questões e proposta de redação.

Essa é a segunda edição do exame realizada durante a pandemia. "Para este ano, não esperamos grandes mudanças. É fato que todo ano tem algumas pequenas variações. Às vezes uma prova mais fácil, outra mais difícil, mas não é uma mudança drástica. Acreditamos que o Enem não deva ter mudanças estruturais", avalia o diretor de ensino do Grupo Bernoulli, Rommel Domingos.

Para Domingos, entretanto, parte dos alunos pode enfrentar mais dificuldades devido à preparação de forma remota. "A preparação do aluno foi afetada desde o ano passado, em função da pandemia. Grande parte deles se adaptou bem ao ensino remoto, conseguiu desenvolver e crescer apesar das dificuldades. Está agora se preparando para o Enem e não deve ter perda no seu rendimento. Mas é fato que há um grupo de alunos que não se adaptou e está tendo dificuldade de administrar o seu tempo e estudar. Em particular, os alunos mais pobres, que têm menos espaços em sua residência, não têm bons equipamentos nema internet de qualidade."

Outro que não acredita em mudança no nível das provas é o diretor da Associação Pré-PUC e UFMG Pré-Vestibular, Richard Thuin. "A maior alteração seria a prova on-line. As duas vão ser no mesmo dia este ano. Quanto ao conteúdo não vejo diferença não." Ele também destaca que a pandemia dificultou a preparação dos alunos, principalmente os mais carentes.

"Eles tiveram aula online, alguns conseguiram acompanhar, outros não. Em função disso, estamos formando muitos alunos no EJA (Educação de Jovens e Adultos). Principalmente alunos de 3° ano que não concluíram (os estudos) na escola pública e estão buscando o curso EJA. Com a pandemia, os pais de alguns alunos perderam o emprego e os filhos precisaram buscar atividades para ajudar no sustento da casa. Com isso, alguns pararam de estudar e resolveram terminar o curso com o EJA, por ser mais rápido. E ele habilita o aluno a fazer as provas do Enem", explica.

Mas, para Rommel Domingos, existe um fator positivo para a edição deste ano. "A ansiedade é menor. Este ano a pandemia continua. O retorno do pré-vestibular já aconteceu e do colégio ainda não, estamos aguardando. É um retorno muito comedido, muitos alunos ainda continuam em casa. Mas a ansiedade este ano é muito menor que a do ano passado. O ambiente não está tão pesado", acredita.

Para ele, o fato dos cursinhos já terem retomado as aulas presenciais pode ajudar nesse problema. "Tem a questão do pré-vestibular já ter voltado. Então, vêm para a aula presencial justamente os que precisam mais. O aluno que já está mais adaptado, continua estudando em casa. E o aluno que está sofrendo mais e não conseguiu se adaptar, acaba entrando no rodízio e sendo beneficiado. Foi importantíssima a volta ao presencial, nesse caso."

Richard Thuin acredita que as aulas on-line exigem um esforço maior dos alunos. "Na aula online, os alunos ficam mais tímidos e não perguntam. As aulas ficam gravadas. O curso remoto precisa de muita dedicação. Com as vídeoaulas, apesar de o aluno ter mais carga de conhecimento e mais tempo, é mais superficial. O aluno não aprende o contexto todo. Esse aluno vai precisar de mais interesse e, alguns estudantes nessa idade têm pouco interesse." Apesar disso, ele afirma que a aprovação a última edição do Enem não foi afetada. "Em 2020, diversos alunos de escolas públicas continuaram as aulas mesmo a distância e tivemos uma aprovação muito expressiva, de 86

Ana Beatriz Gontijo tem 19 anos e vai fazer o Enem pela segunda vez. Ela se formou em 2020 e pretende estudar medicina. No último Enem, as incertezas sobre o processo a deixaram angustiada.

"No início deste ano, foi a primeira vez que fiz (o Enem) valendo, não como treineira. No início, a adaptação às aulas on-line foi muito difícil. E a confusão toda de fazer enquete, disseram que seria uma data, depois foi em outra. Só foi adiando. Isso foi muito angustiante, a gente não tinha nenhum cronograma para seguir e planejar".

Ela acredita que o impasse sobre a realização do exame pode ter atrapalhado seu desempenho. "Não posso colocar a culpa 100% nisso, mas acho que prejudicou sim. Foi uma prova muito diferente em relação ao conteúdo. Por exemplo, história do Brasil, não caiu praticamente nenhuma questão. E os professores nos prepararam de acordo com as experiências anteriores do Enem."

A estudante conta que, este ano, a preparação está sendo mais tranquila. "Como a gente já teve a experiência do on-line de 2020, agora está sendo um pouco mais fácil. Já peguei o ritmo, o jeito de estudar em casa. Inclusive, estou no pré-vestibular em que as aulas presenciais já voltaram (no modelo híbrido), mas nem estou indo porque já me acostumei à rotina de estudar em casa."

Outro ponto positivo, segundo Ana Beatriz, é o exame já ter data definida. No ano passado, a incerteza persistiu até o fim, inclusive com medidas judiciais para suspender o exame nacional. Já Wanderson dos Santos, de 24 anos, vive uma situação diferente. Ele vai tentar o Enem pela terceira vez, sempre estudou em escola pública municipal e terminou o ensino médio em 2015. Também quer estudar medicina, mas não conseguiu se adaptar ao ensino remoto. "Foi uma experiência muito ruim por causa da distração. Na aula presencial, a gente tem um compromisso maior. Na aula on-line não, como ela é gravada, você começa a fazer outras coisas e quando vê está com um monte de matéria acumulada".

D.2 Text 4 in English

Registration for the 2021 edition of the National High School Exam (Enem) begins today, open until July 14th. The tests are scheduled for the 21st and 28th of November, in what will be the second edition of Enem since the beginning of the COVID-19 pandemic. Experts estimate that there will be no major changes in terms of content and are betting on reducing tension compared to the last edition, with students now more accustomed to distance learning and even being able to access face-to-face classes in preparatory courses. They remember, however, that this adaptation is uneven, especially for the poor. A more defined calendar also contributes to better expectations, in the view of students who faced uncertainty in the last version of Enem, held this year after postponements.

To register, candidates must access the participant page and choose between taking the printed or digital test. This year, the National Institute of Educational Studies and Research Anísio Teixeira (Inep), responsible for carrying out the Enem, will offer more than 101 thousand registrations for the digital version of the test. Only students who have already completed high school or will complete it in 2021 can opt for this model. The printed and digital tests will be taken on the same date and, therefore, will have the same questions and writing proposal.

This is the second edition of the exam carried out during the pandemic. "For this year, we don't expect any major changes. It is a fact that every year there are some small variations. Sometimes a test is easier, sometimes more difficult, but it's not a drastic change. We believe that Enem should not have structural changes", assesses the teaching director of the Bernoulli Group, Rommel Domingos.

For Domingos, however, some students may face more difficulties due to remote preparation. "Student preparation has been affected since last year, due to the pandemic. Most of them adapted well to remote teaching and managed to develop and grow despite the difficulties. You are now preparing for the Enem and should not have any loss in your income. But it is a fact that there is a group of students who have not adapted and are having difficulty managing their time and

studying. In particular, the poorest students, who have less space in their homes, do not have good equipment or quality internet."

Another person who does not believe in changing the level of tests is the director of the Associação Pré-PUC and UFMG Pré-Vestibular, Richard Thuin. "The biggest change would be the online test. The two will be on the same day this year. As for the content, I don't see any difference." He also highlights that the pandemic made it difficult for students to prepare, especially those most in need.

"They had online classes, some were able to follow it, others were not. As a result, we are training many students in EJA (Youth and Adult Education). Mainly 3rd year students who did not complete (their studies) at public school and are looking for the EJA course. With the pandemic, the parents of some students lost their jobs and their children had to look for activities to help support the household. As a result, some stopped studying and decided to finish the course with EJA, as it was faster. And it enables the student to take the Enem exams", he explains.

But, for Rommel Domingos, there is a positive factor for this year's edition. "The anxiety is less. This year the pandemic continues. The return of the pre-university exam has already happened and the college has not yet, we are waiting. It is a very measured return, many students are still at home. But the anxiety this year is much lower than last year. The environment is not so heavy", he believes.

For him, the fact that the courses have already resumed face-to-face classes can help with this problem. "There is the issue of the pre-university exam having already returned. So, those who need it most come to the in-person class. The student who is already more adapted continues studying at home. And the student who is suffering the most and has not been able to adapt ends up entering the rotation and benefiting. It was very important to return to in-person, in this case."

Richard Thuin believes that online classes require greater effort from students. "In online classes, students are more shy and don't ask questions. Classes are recorded. The remote course requires a lot of dedication. With video classes, although the student has more knowledge and more time, it is more superficial. The student does not learn the entire context. This student will need more interest, and some students at this age have little interest." Despite this, he states that the approval of the latest edition of Enem was not affected. "In 2020, several public school students continued their classes even remotely and we had a very impressive approval rate of 86%."

Ana Beatriz Gontijo is 19 years old and will take the Enem for the second time. She graduated in 2020 and plans to study medicine. In the last Enem, the uncertainties about the process left her distressed. "At the beginning of this year, it was the first time I took (the Enem) for real, not as a trainer. At first, adapting to online classes was very difficult. And the whole mess of doing a survey, they said it would be on one date, then it was on another. It just kept postponing. This was very distressing, we had no schedule to follow and plan."

She believes that the impasse over the exam may have hindered her performance. "I can't blame it 100% on that, but I think it did hurt. It was a very different test in terms of content. For example, the history of Brazil, practically no questions dropped. And the teachers prepared us according to previous Enem experiences."

The student says that, this year, preparation is going more smoothly. "As we already had the online experience of 2020, now it is a little easier. I've already got the rhythm, the hang of studying at home. In fact, I'm in the pre-university exam where face-to-face classes have already returned (in the hybrid model), but I'm not even going because I've already gotten used to the routine of studying at home."

Another positive point, according to Ana Beatriz, is that the exam already has a defined date. Last year, uncertainty persisted until the end, including legal measures to suspend the national exam. Wanderson dos Santos, 24 years old, lives a different situation. He will try Enem for the

third time, he always studied at a municipal public school and finished high school in 2015. He also wants to study medicine, but was unable to adapt to remote teaching. “It was a very bad experience because of the distraction. In face-to-face classes, we have a greater commitment. Not in the online class, as it is recorded, you start doing other things and when you see it you have a lot of material accumulated”.

D.3 Questions for Text 4

(1) What dates was ENEM 2021 scheduled for?
(2) What challenges did poorer students face in preparing for the ENEM during the pandemic?
(3) What course does student Ana Beatriz want to take?

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